USER INTERFACE SPECIFICATION

for

Boomaga-IPP

Version 0.2.0
(Draft Baseline)

Prepared by Gary S. Martin
with AI tool assistance

***Boomaga-IPP* Community**

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1. Introduction

1.1. Purpose

The purpose of the *Boomaga-IPP Virtual Printer* system is to replace the venerable, but now orphaned *Boomaga*[1] virtual printer (Sokolov et al., 2013b; Sokolov et al., 2013a) with a free, open-source, memory-safe, and type-safe virtual printer system for current *Linux*® computer systems. The *Boomaga-IPP Virtual Printer* system is intended to conform to the current standards of the IEEE ISTO Printer Working Group (PWG): Internet Printing Protocol (IPP) Version 2.0 (IPP/2.0) and *IPP Everywhere*™. *Boomaga-IPP* is targeted at *systemd*-managed *Linux*® systems with *Wayland* display environments. It is to be written completely in the *RUST*® programming language and is to use the *Xilem* GUI toolkit.

1.2. Document Conventions

The format of this document is adapted from the *LaTeX template for IEEE Software Requirements Specifications* (Eisenbarth, 2014-09-06/2026). M. Eisenbarth's template is partially derived from the code of Yiannis Lazarides and the template of Karl E. Wiegers. The document template is based upon and employs the document conventions of *IEEE 830-1998, Recommended Practice for Software Requirements Specifications* (IEEE Computer Society, 1998). *IEEE 830-1998* is a standard of the IEEE Computer Society, and is not freely available. While it was used in the development of the *LaTeX template*, it was not referenced directly in the production of this document.

The use of the word "shall" in this document connotes a requirement. Each requirement herein is numbered as a Level 2 (*L2*) *UI* Requirement and is sequentially numbered based on its order of occurrence in the initial draft of the document. To ensure traceability, requirements will not be renumbered if the document changes. Any requirements that are deleted will result in gaps in the numbering and any new requirements added will have higher numbers than previously numbered requirements regardless of their place in the specification or their precedence.

All requirements specified in this user interface specification (UIS) are derived from the *Boomaga-IPP* software requirements specification (SRS) (Martin and Boomaga-IPP Project Team, 2026-03-12/2026b). Within this UIS, the priorities of higher-level parent requirements are assumed to be inherited by their detailed child requirements unless otherwise specified. In the event of any discrepancy between the SRS and the UIS, the SRS shall be the governing document. *(L2-UI-001)*

1.3. Intended Audience and Reading Suggestions

This UIS is intended for use by any and all members of the *Boomaga-IPP* Project Team, including the project manager, software developers, alpha and beta testers, and documentation writers. This UIS es-

[1]Boomaga is a portmanteau of "BOOklet MAnager"

tablishes all functional and nonfunctional requirements governing the development of the *Boomaga-IPP Virtual Printer* system user interface (UI). Any and all *Boomaga-IPP* team members should familiarize themselves with the entirety of this specification. Technically inclined *Boomaga-IPP* users may also wish to familiarize themselves with any sections of this UIS that interest them.

1.4. Overview

The objective of the *Boomaga-IPP* project is to deliver a fully-functional, easily-installable, free, and open-source virtual printer system for current *Linux*® computer systems. This virtual printer system is intended to provide the same or greater functionality as the original *Boomaga* virtual printer system with greater reliability and the same or greater ease of use. This virtual printer system shall be comprised of both software and documentation. *(L2-UI-002)* The *Boomaga-IPP* project team intends to maintain the software and documentation for at least five years from initial release or until the system is replaced by a virtual printer system providing the same or greater functionality for the majority of common *Linux*® desktop environments.

1.5. Basic Goals

This UIS establishes the basic requirements and development processes that shall be used to produce the *Boomaga-IPP* GUI. *(L2-UI-003)* These development processes shall require the use of any software tools specified in the SRS or in this UIS. *(L2-UI-004)* This UIS shall conform to the requirements of the *Boomaga-IPP* SRS. *(L2-UI-005)* The SRS is available from the *Boomaga-IPP* Project Repository (Martin & Boomaga-IPP Project Team, 2026-03-03/2026a).

The *Boomaga-IPP* GUI shall be designed to provide a simple and intuitive graphical interface to its user. *(L2-UI-006)* It shall provide the primary functions enumerated in Table 1.1. *(L2-UI-007)* All provided GUI functions shall be available via keyboard commands as well as via graphical controls and displays. *(L2-UI-008)* The major common component libraries that the *Boomaga-IPP* GUI depends upon are enumerated in Table 1.2.

Table 1.1.: *Boomaga-IPP* GUI Primary Functions.

Boomaga-IPP GUI shall:

1. Preview one or more files to be printed as it or they will appear coming out of the printer,
2. Rotate selected pages clockwise or counterclockwise, if desired,
3. Delete selected pages from a set of documents, if desired,
4. Add blank pages at selected locations to a set of documents, if desired,
5. Print the edited document file in half-page booklet format, if desired,
6. Print the edited document file with a user-selected number of pages per sheet, if booklet format is not desired,
7. Print the final output file to any suitable *CUPS* printer selected by the user.

(L2-UI-007)

Table 1.2.: *Boomaga-IPP* GUI Common Library Dependencies.

Library	Author	Minimum Version	Provided Function
Linebender Xilem GUI Framework	Linebender Organization	0.4.0	Graphical Design
Poppler-rs	Dröge et al.	0.26.0	PDF Rendering
qpdf	Pankratov	0.3.5	PDF Manipulation

1.6. Using the Program — The Big Picture

Boomaga-IPP is designed to perform the functions of previewing, imposing (laying out), and printing a set of documents in a single GUI session. Consequently, its single GUI use case is *Preview, Imposition (Lay Out), and Print a Set of Documents*. The textual form of the use case is shown as Table 1.3. The Unified Modeling Language (UML) use case diagram is shown as Figure 1.1. Additionally, the UML component diagram, class diagram, sequence diagram, and use case diagram for the *Boomaga-IPP Virtual Printer* system are included in Appendix C, Analysis Models, as Figures C.1, C.2, C.3, and C.4, respectively.

Table 1.3.: *Boomaga-IPP* Use Case.

Preview, Impose (Lay Out), and Print a Set of Documents Goal Level: Sea Level
MAIN SUCCESS SCENARIO: 1. Print one or more documents to *Boomaga-IPP*. 2. Preview the resulting print job or jobs. 3. Manipulate Pages as required: a) Rotate pages as required. b) Delete pages as required. c) Add Blank Pages as required. 4. Select desired imposition (layout). 5. Select desired downstream printer. 6. Print job or jobs.
EXTENSIONS: 6a: Select desired PDF output file: .1 Select “Print to File (PDF)” or “CUPS-PDF.” (if CUPS-PDF is installed) .2 Select Filename and Folder from printer dialog.

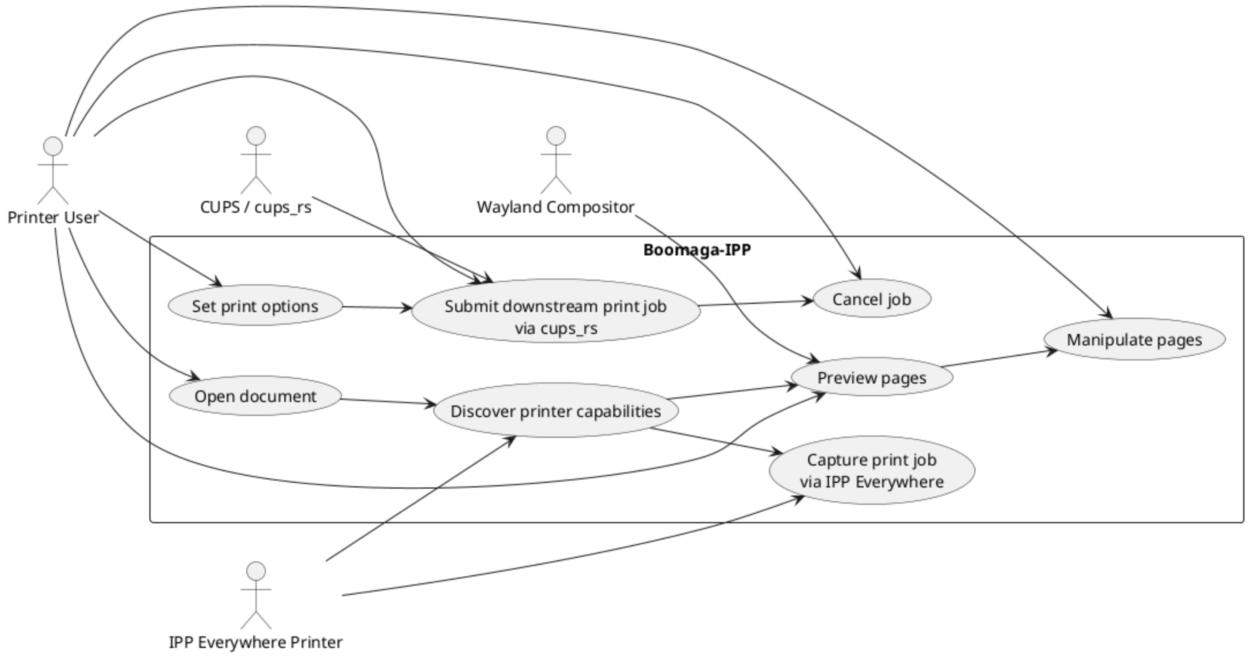

Figure 1.1.: *Boomaga-IPP* Use Case.

1.7. Common Use Scenarios

1.7.1. Previewing a Set of Documents

In the *Boomaga-IPP* GUI display window, any print job or sequential set of print jobs captured by the *Boomaga-IPP Virtual Printer* shall be displayed one full imposed page at a time (for N-Up imposition) or one full printer spread at a time (for booklet imposition). *(L2-UI-009)* The user shall be able to view the current imposition (layout) of any desired imposed page or printer spread in the captured print job (or sequential set of captured print jobs) by scrolling to it via the user's choice of GUI commands, keyboard commands, or mouse commands. *(L2-UI-195)* The document preview presented to the user shall be updated immediately as the user makes changes to the desired output state through the *Boomaga-IPP GUI Application*. *(L2-UI-237)*

1.7.2. Manipulating Pages

The user shall be able to manipulate selected individual document pages or groups of document pages, including entire print jobs, through the use of GUI commands, keyboard commands, or mouse commands. *(L2-UI-196)* Commanded manipulations shall include: 1) rotating indivual document pages or groups of document pages, including entire print jobs, clockwise or counterclockwise, 90 degrees at a time; 2) deleting document pages or groups of document pages, including entire print jobs; and 3) inserting blank pages before or after a selected document page or group of document pages. *(L2-UI-010)* The user shall be able to reorder document pages within a job and reorder jobs within the Job_Store output sequence, using drag-and-drop and/or equivalent keyboard commands. *(L2-UI-235)* Page manipulations shall update the output sequence state object only (never the underlying Job_Store), consistent with *(L2-UI-047)*. *(L2-UI-236)*

Rotating Pages

The user shall be able to command clockwise or counterclockwise 90 degree at a time rotation of individual document pages or groups of document pages, including entire print jobs. *(L2-UI-011)* Page rotation commands shall be available through the user's choice of GUI commands, keyboard commands, or mouse commands. *(L2-UI-012)*

Deleting Pages

The user shall be able to command the deletion of individual document pages or groups of document pages, including entire print jobs. *(L2-UI-013)* Page deletion commands shall be available through the user's choice of GUI commands, keyboard commands, or mouse commands. *(L2-UI-014)*

Inserting Blank Pages

The user shall be able to command the insertion of blank document pages before or after selected individual document pages or groups of document pages. *(L2-UI-015)* Page insertion commands shall be available through the user's choice of GUI commands, keyboard commands, or mouse commands. *(L2-UI-016)*

1.7.3. Selecting Desired Imposition (Layout)

The user shall be able to select the desired imposition (layout) of the *Boomaga-IPP* output. *(L2-UI-017)* Imposition commands shall be available through the user's choice of GUI commands, keyboard commands, or mouse commands. *(L2-UI-018)* Imposition (Layout) commands shall accommodate both portrait and landscape printing orientations. *(L2-UI-019)*

Imposition of a User-Selected Number of Pages Per Sheet (N-up)

The user shall be able to command that the *Boomaga-IPP* output be imposed into the following formats: one document page per imposed page (1-up), two document pages per imposed page (2-up), four document pages per imposed page (4-up), and eight document pages per imposed page (8-up). *(L2-UI-024)* For 4-up and 8-up impositions, the user shall be able to select horizontal (row-major) or vertical (column-major) page ordering. *(L2-UI-232)* The JOB_STORE_PDF_PREVIEW shall reflect the selected page ordering immediately. *(L2-UI-233)* These N-up page imposition commands shall accommodate both portrait and landscape printing orientations. *(L2-UI-025)* The N-up imposition commands shall be available through the user's choice of GUI commands, keyboard commands, or mouse commands. *(L2-UI-026)* The user shall be able to select simplex (single-sided / one imposed page per sheet) or duplex (double-sided / two imposed pages per sheet) printing of the output. *(L2-UI-027)* If the selected Downstream Printer is not duplex capable, the *Boomaga-IPP* IMPOSITION_ENGINE shall arrange the output for manual duplex printing. *(L2-UI-028)* When the user selects N-up Imposition, the *Boomaga-IPP* JOB_STORE_PDF_PREVIEW Window shall display the single imposed page selected by the user. *(L2-UI-197)*

Imposition of Half-Page Booklets

The user shall be able to command imposition of the *Boomaga-IPP* output into a four pages per sheet booklet format. *(L2-UI-020)* Booklet layout commands shall accommodate both portrait and landscape printing orientations. *(L2-UI-021)* Booklet imposition commands shall be available through the user's choice of GUI commands, keyboard commands, or mouse commands. *(L2-UI-022)* Booklet format shall assume duplex double-sided/two imposed pages per sheet) printing. *(L2-UI-023)* When the user selects Booklet Imposition, the *Boomaga-IPP* Job_Store_PDF_Preview Window shall display the printer spread selected by the user. *(L2-UI-198)*

Configuration of Printer spread Settings

The user shall be able to set the top, bottom, left, and right page margins of the imposed output, in millimeters. *(L2-UI-205)* The user shall be able to set an internal (binding gutter) margin, in millimeters. *(L2-UI-206)* User shall be able to "Reset to default," that is restore all margin values to their defaults. *(L2-UI-207)* The user shall be able to enable or disable drawing a border around each imposed page (default disabled). *(L2-UI-208)*

1.7.4. Selecting Desired Downstream Printer and Printing

The user shall be able to select any suitable *CUPS* printer as the downstream printer through integration of the GUI with the host system's *CUPS* server. *(L2-UI-029)*

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2. System Concept[1]

2.1. Product Perspective

Boomaga is a virtual printer for viewing a document before printing it out to either a physical printer or another virtual printer on *Linux*®-based computer systems. It is also capable of printing booklets and other arrangements of multiples pages per sheet, and of combining multiple documents into a single print job. *Boomaga* is an open source software project mainly distributed under the GPLv2 license, although some files are distributed under the LGPLv2+ license. The *Boomaga* project began in 2012 and the first software release was Dec 14, 2013. The final release of *Boomaga*, version 3.0.0, was on March 13, 2019. On February 20, 2022, a patch to provide compatibility with *GhostScript*® V9.54 was merged with the *Boomaga* source code, but no release was issued. *Boomaga* has been unmaintained since that date.

Basic compatibility with *Wayland* was added to *Boomaga* in v3.0.0 in 2019, but *Boomaga* was primarily designed to work under *X11* graphical environments (which either have been or are actively being replaced by *Wayland* environments in popular *Linux*® distributions). In addition, the *D-Bus* mechanism used by *Boomaga* for communication between its user-space GUI and the host computer's root-level *CUPS* print driver has proven unreliable for many users, preventing the GUI from automatically starting when the user prints a file to *Boomaga*. A number of recent minor issues have also arisen from incompatibilities between *Boomaga* and regular version updates in the tool chain and library packages used to build and install it.

Boomaga-IPP was conceived in late 2025 as a replacement for *Boomaga*. Development of *Boomaga-IPP* began in February 2026 with the assistance of newly available artificial intelligence coding tools including *CLAUDE*® *Code*, *opencode*®, *Perplexity.ai*, and several open-source, locally-run large language models (LLMs) developed for coding.

Boomaga-IPP will not reuse any *Boomaga* code, but is to be written from scratch in a high-reliability language to provide type safety and memory safety. *Ada* was briefly considered (based on the project manager's experience with it), but was rejected due the lack of existing *Ada* language bindings to the libraries needed by *Boomaga-IPP*. It was decided that *Boomaga-IPP* will be written in the *RUST*® programming language, as the needed *RUST*® language bindings are readily available.

Boomaga-IPP is targeted at *Linux*® systems employing *Wayland* display architectures and will use an *IPP Everywhere*™ client in place of a printer driver. *Boomaga-IPP* is intended to provide the same or greater functionality as the original *Boomaga*.

Boomaga-IPP will support the *IPP Everywhere*™ mandatory document formats: PDF (application/pdf), PWG Raster (image/pwg-raster), and JPEG (image/jpeg). Since these do not include *POSTSCRIPT*®, *Boomaga-IPP* will not continue *Boomaga*'s support of *POSTSCRIPT*®. Consequently, *Boomaga-IPP* will use the combination of *Poppler* and *qpdf* to replace the remaining required functions performed by *GhostScript*® in *Boomaga*.

[1] For the sake of completeness of the documents, this chapter is included as the second chapter of both the Software Requirements Specification and the User Interface Specification.

2.2. Product Functions

The top-level functional requirements of the *Boomaga-IPP Virtual Printer* system are allocated to the following major software components:

***Boomaga-IPP* Capture_Daemon** A daemon that captures input print jobs through an *IPP Everywhere*™ print service and adds them to the *Boomaga-IPP* Job_Store.

***Boomaga-IPP* GUI** A desktop application launched by and operating in cooperation with the *Boomaga-IPP* Imposition_Engine. The *Boomaga-IPP GUI Application* allows the user to select the desired imposition settings, preview the output PDF (print job), select the desired downstream printer, select the desired downstream printer settings, and request the printing of the output print job.

***Boomaga-IPP* Imposition_Engine** A daemon that constructs the output print job by imposing the input print job(s) according to the imposition settings and document page manipulation requests received from the user through the *Boomaga-IPP GUI Application*. The Imposition_Engine also sends the output print job to the GUI Job_Store_PDF_Preview feature and the *Boomaga-IPP* Downstream_Printer in response to *Boomaga-IPP Virtual Printer* requests.

***Boomaga-IPP* Downstream_Printer** A daemon that manages the interface between the *Boomaga-IPP Virtual Printer* and the host computer's *CUPS* server, enabling the *Boomaga-IPP* user, through the *Boomaga-IPP GUI Application*, to select a *CUPS*-backed printer, change printer settings of the selected printer, and submit the output print job to the selected downstream printer. The downstream *CUPS* printer may be a physical printer or another virtual printer such as *CUPS-PDF*.

A Unified Modeling Language (UML) component diagram of these software components is displayed below in Figure 2.1. Additionally, the UML component diagram, class diagram, sequence diagram, and use case diagram for the *Boomaga-IPP Virtual Printer* system are included in Appendix C, Analysis Models, as Figures C.1, C.2, C.3, and C.4, respectively.

1. The *Boomaga-IPP* Capture_Daemon will allow *Boomaga-IPP Virtual Printer* users to print through the host computer's *CUPS* server to the *Boomaga-IPP Virtual Printer* via a provided *IPP Everywhere*™ client. The *IPP Everywhere*™ client will:

   a) Interface through *CUPS*,

   b) Queue the printed file to the virtual printer,

   c) Provide document merging and batch printing, and

   d) Launch the *Boomaga-IPP* GUI application on the user's desktop.

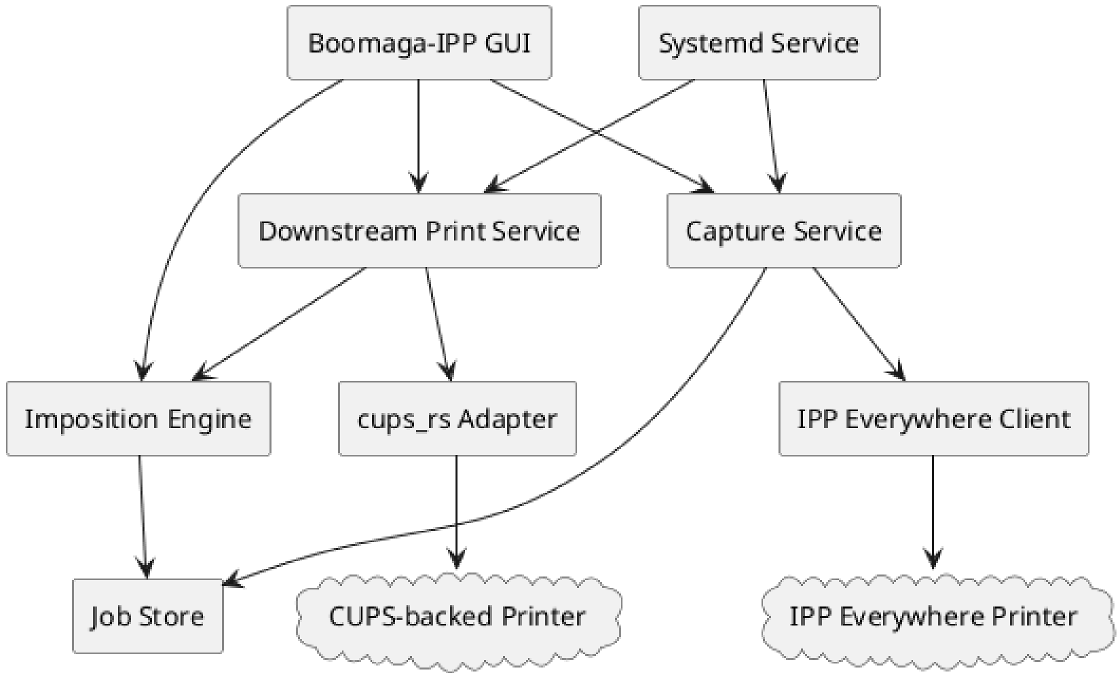

Figure 2.1.: *Boomaga-IPP Virtual Printer* System Component Diagram.

2. The *Boomaga-IPP Virtual Printer* will present a GUI to the user once an input job is received by the CAPTURE_DAEMON and added to the IMPOSITION_ENGINE JOB_STORE. The *Boomaga-IPP GUI Application* will:

   a) Preview prints to the user prior to physical printing;

   b) Allow the user to select any other *CUPS* printer for the virtual printer output;

   c) Allow the user to print booklets with automatic page arrangement;

   d) Allow the user to print multiple pages per sheet (1, 2, 4, 8 pages per sheet) with horizontal and vertical page ordering;

   e) Allow the user to manipulate pages by rotating pages, deleting pages, adding blank pages, and reordering pages;

   f) Allow the user to create and manage sub-booklets;

   g) Provide duplex printing for both duplex-capable and manual duplex printers; and

   h) Send the virtual printer output to the downstream *CUPS* printer in the form selected by the user via the *Boomaga-IPP* DOWNSTREAM_PRINTER daemon.

3. The *Boomaga-IPP Virtual Printer* will provide inter-process communication between the *Boomaga-IPP* software components described above as required to execute the functions of the *Boomaga-IPP Virtual Printer*, including printing the user's desired output to the downstream *CUPS* printer.

2.3. User Classes and Characteristics

A single class of *Boomaga-IPP Virtual Printer* users is envisioned. All users are expected to install the *Boomaga-IPP Virtual Printer* using the provided package manager of their *Linux*® distribution whenever it is available through that channel. After installation, users will employ *Boomaga-IPP* by selecting it from the menu provided by *CUPS* to the various applications installed on the users' systems and print a single file or a series of files to the *Boomaga-IPP Virtual Printer*. Printing a file to the *Boomaga-IPP Virtual Printer* will cause the *Boomaga-IPP GUI Application* to launch automatically on the user's desktop. Using the *Boomaga-IPP GUI Application*, users will be able to:

1. Manipulate pages as desired,
2. Select the desired imposition format,
3. Select the downstream *CUPS* printer and change its settings, and
4. Print the output to the selected downstream printer.

The *Boomaga-IPP* user experience is expected to be very similar to the experience currently provided by *Boomaga*.

2.4. Operating Environment

Boomaga-IPP's initial development will take place on an x86-64 computer system running current versions of the *ARCH LINUX*™ distribution, *Wayland*, and *KDE Plasma*. The x86-64 architecture is the primary target architecture for the *Boomaga-IPP Virtual Printer* system. The software environment necessary to support *Boomaga-IPP* is identified in Table 2.1 (including the versions in use as of the date of this document). It is expected that any ARM-64 system capable of hosting a robust modern *Linux*® distribution and *Wayland* display environment should also be capable of compiling the *Boomaga-IPP* system. The *Boomaga-IPP* team intends to test the system on an ARM-64 architecture computer system at a later time.

2.5. Design and Implementation Constraints

All *Boomaga-IPP* software will be released under the GNU General Public License, version 3 (GPLv3). All *Boomaga-IPP* software will be written in the *RUST*® programming language. The *Boomaga-IPP Virtual Printer* system will be initially targeted at modern x86-64 computer systems running a *Linux*® version 7 or later kernel, the *Wayland* display architecture, and *systemd* for system administration. The *Boomaga-IPP* GUI will be based on the *Linebender Organization Xilem* GUI toolkit (Linebender Organization, nodate-a, nodate-b, 2022-11-16/2026). *Boomaga-IPP* will use *Poppler-rs* (Dröge et al., 2026; Høgsberg & The Poppler Project, nodate) for document rendering. *Boomaga-IPP* will use the *zbus_systemd* (Bruno & Khan, 2026) library for inter-process communication (IPC) with *systemd*.

Table 2.1.: *Boomaga-IPP* Software Environment.

Core System Software		Rust crates			
Required Software	Version	Required Software	Version	Required Software	Version
Linux kernel	6.18.16	GUI		Logging	
glibc	2.43	xilem	0.4	tracing	0.1
Wayland	1.24.0	kurbo	0.11	tracing-subscriber	0.3
systemd	259.3	winit	0.3		
poppler-glib	26.0.2	cairo-rs	0.18		
qpdf	12.3.2				
		Document Rendering		Document Manipulation	
		poppler-rs	0.26.0	qpdf	0.3.5
		libc	0.2		
		IPC		Configuration	
		zbus_systemd	0.26100.0	config	0.14
				directories	5.0
		Async Runtime		Time	
		tokio	1.35	chrono	0.4
		async-trait	0.1	URL	
				url	2.5
		Serialization		Temporary Files	
		serde	1.0	tempfile	3.1
		serde_json	1.0		
		Error Handling		UUID	
		thiserror	1.0	uuid	1.0
		anyhow	1.0		

2.6. User Documentation

The primary user documentation will be the README.md file on the *Boomaga-IPP* github archive (Martin & Boomaga-IPP Project Team, 2026-03-03/2026a). At a later time, a frequently asked questions (FAQ) page will be added to the wiki on the *Boomaga-IPP* github archive. Any supplemental documentation developed will be hosted there, as well.

2.7. Assumptions and Dependencies

Successful implementation of *Boomaga-IPP* relies upon two immature technologies: artificial intelligence (AI) coding assistance and the *Linebender* *Xilem* GUI toolkit. The project initiator had virtually no knowledge of the *RUST*® programming language at the start of the project. As a result, a heavy reliance is being placed on AI assistance. *CLAUDE*® *Code*, *opencode*®, and *Perplexity.ai* were used to establish the initial project requirements and system architecture. Cost limitations soon dictated the use of the *GLM-4.7-flash* local LLM together with *CLAUDE*® *Code* to author the prototype code. The code was further developed with the *Meta-Llama-3.1-8B-Instruct* model because of its superior speed. It is expected that additional AI tools will be used as they emerge and mature. A number of local and cloud-based options are being assessed for code review and testing. The maturity and cost of the various options will have a substantial impact on the quality and functionality of the final code and may well determine whether or not the *Boomaga-IPP* project produces a usable virtual printer implementation.

The selection of the *Linebender Organization*'s *Xilem* GUI toolkit for use in the development of the *Boomaga-IPP* GUI was made because it was judged to be the best option for producing a maintainable *RUST*® based system over the long term. The original choice was the *Linebender* *Druid* toolkit, also written entirely in *RUST*®. However, *Druid* is deprecated and no longer under development by the *Linebender Organization*. As one of the goals of this project is to produce a virtual printer system that is maintainable over a decade or more, it was decided to move from *Druid* to *Xilem* even though *Xilem* is alpha level code as of the date of this document. Consequently, the maintainability of the *Boomaga-IPP* system depends upon the *Linebender Organization* continuing its development of the *Xilem* toolkit through beta testing and into a stable production framework.

3. User Interface Features

The *Boomaga-IPP* GUI shall emulate the *Boomaga* GUI with differences as described in the *UI* feature sections that follow. *(L2-UI-030)* An annotated screen shot of the *Boomaga* UI is shown in Figure 3.1. The colored boxes and accompanying text annotations indicate the *Boomaga* UI features assuming the architecture adopted for *Boomaga-IPP* and described below.

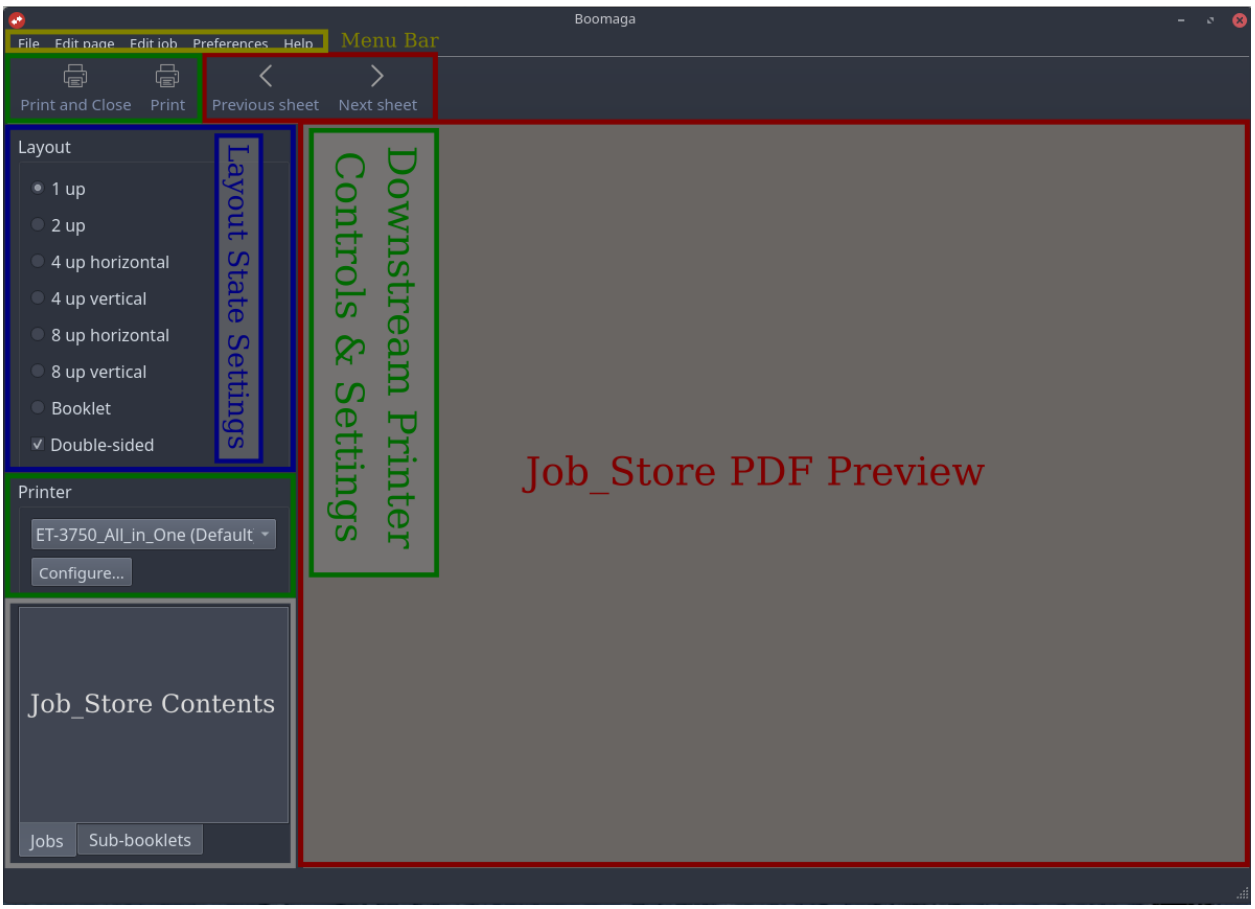

Figure 3.1.: *Boomaga* Screenshot with *Boomaga-IPP UI* Feature Annotations.

The UI features are listed in the following sections in the order of priority to be given them in the *Boomaga-IPP* UI. The first and primary feature shall be the JOB_STORE_CONTENTS_LISTING display (annotated in grey). *(L2-UI-031)* The next feature in priority order shall be the JOB_STORE_PDF_PREVIEW window and associated page view controls (annotated in red). *(L2-UI-032)* Third shall be the Commanded Imposition (Layout) State Setting Controls (annotated in blue). *(L2-UI-033)* Fourth shall be the Downstream Printer Controls and Settings (annotated in green). *(L2-UI-034)* After these features in priority shall come the standard Menu Bar (annotated in mustard), Content Specific Menus (not shown here), and Window Decoration (partially shown, not annotated) UI elements. *(L2-UI-035)*

3.1. Feature 1: Job_Store_Contents_Listing

3.1.1. Description and Priority

The Job_Store_Contents_Listing shall have two user selectable display modes: "Jobs" and "Sub-booklets."*(L2-UI-036)* The default mode shall be the "Jobs" mode. *(L2-UI-037)* In the "Jobs" mode, the content listing display shall show, in order, all the jobs received through the *IPP Everywhere*™ Client and added to the Job_Store. *(L2-UI-038)* In the "Sub-booklets" mode, the content listing display shall show, in order, all the Sub-booklets as they will be built upon a print command from the user using the current Job_Store, current output sequence state object, and current Sub-booklet settings. *(L2-UI-039)*

3.1.2. Stimulus/Response Sequences

The *Boomaga-IPP* GUI shall permit the user to select either the "Jobs" tab or the "Sub-booklets" tab. *(L2-UI-040)* Selecting the "Jobs" tab shall switch the Job_Store Content Listing to "Jobs" mode. *(L2-UI-041)* Selecting the "Sub-booklets" tab shall switch the Job_Store Content Listing to the "Sub-booklets" mode. *(L2-UI-042)*

In "Jobs" mode, the GUI shall permit the the user to select one or more of the jobs displayed in the Job_Store content listing with the mouse. *(L2-UI-043)* The user shall be able to command deletion of selected jobs using the Menu Bar, a context menu via mouse command, and the Delete key of the keyboard. *(L2-UI-044)*

In "Sub-booklets" mode, the GUI shall permit the user to select one or more of the Sub-booklets displayed in the Job_Store content listing with the mouse. *(L2-UI-045)* The user shall be able to command deletion of selected sub-booklets using the Menu Bar, a context menu via mouse command, and the Delete key of the keyboard. *(L2-UI-046)*

3.1.3. Functional Requirements

The GUI shall not modify the contents of the Job_Store, but shall maintain a state object which shall fully describe the output sequence of pages of the Job_Store as they are to be previewed and printed. *(L2-UI-047)* During the initial loading of the Job_Store, both the input sequence state and the output sequence state shall be constructed. *(L2-UI-048)* The output sequence state object shall be identical to the input sequence state object until the user commands a change in the output sequence state. *(L2-UI-049)* The output sequence state object shall be updated as each command to delete a Job or a Sub-booklet is received from the user through the GUI. *(L2-UI-050)* The Job_Store Content listing and the Job_Store_PDF_Preview page shall be updated as commanded by the user immediately following the output sequence user state update. *(L2-UI-051)*

3.2. Feature 2: Job_Store_PDF_Preview

3.2.1. Description and Priority

The Job_Store_PDF_Preview feature shall display a single imposed page when N-up imposition is selected and a single printer spread when Booklet Imposition is selected. *(L2-UI-052)* The page view controls shall permit the user to page forward or backward through the Job_Store one imposed page or printer spread at a time. *(L2-UI-053)* In *Boomaga*, the page view controls are at the top of the Job_Store_PDF_Preview window. In *Boomaga-IPP*, the page view controls shall be centered beneath the Job_Store_PDF_Preview window between it and the standard window decoration. *(L2-UI-054)* When an imposed page or printer spread is first displayed the first document page shall be set as the selected document page. *(L2-UI-199)* The user shall be able to change the selection to any other visible document page with the mouse or keyboard. *(L2-UI-200)* Document rendering for preview shall use *Poppler-rs* (Dröge et al., 2026).

3.2.2. Stimulus/Response Sequences

Pressing the page forward control button shall advance the imposed page or printer spread displayed in the Job_Store_PDF_Preview display by a single imposed page or printer spread. *(L2-UI-055)* Pressing the page back control button shall cause the imposed page or printer spread displayed in the Job_Store_PDF_Preview display to go back by a single imposed page or printer spread. *(L2-UI-056)* Page forward and page back shall also be commanded by use of the mouse scroll wheel and the keyboard arrow keys. *(L2-UI-057)*

3.2.3. Functional Requirements

The GUI shall not modify the contents of the Job_Store, but shall command updates to a state object which shall fully describe the output sequence of the imposed pages or printer spreads of the Job_Store as they are to be previewed and printed. *(L2-UI-058)* The state for each imposed page or printer spread in the sequence shall indicate the commanded placement and orientation of the previewed/printed imposed page / printer spread in the output sequence of the Job_Store. *(L2-UI-059)* The page forward and page back commands shall move the displayed imposed page / printer spread by one in the desired direction through the output sequence of Job_Store pages. *(L2-UI-060)* During the initial loading of the Job_Store, both the input sequence state and the output sequence state shall be constructed. *(L2-UI-061)* The output sequence state object shall be identical to the input sequence state object until the user commands a change in the output state. *(L2-UI-062)* The output sequence state object shall be updated as each command to add, delete, rotate, or impose document pages is received from the user through the GUI. *(L2-UI-063)* The page view shall be updated as commanded by the user immediately following the output sequence user state update. *(L2-UI-064)*

3.3. Feature 3: Primary Imposition (Layout) Controls

3.3.1. Description and Priority

The Primary Imposition (Layout) Controls shall permit the user to specify the "gross" format of the output imposed pages or printer spreads to be sent to *Boomaga-IPP* DOWNSTREAM_PRINTER as specified above in Subsection 1.7.3. *(L2-UI-065)* The Primary Imposition (Layout) Controls shall be placed directly below the Downstream Printer selection and settings controls in the main *Boomaga-IPP GUI Application* window. *(L2-UI-067)* The Primary Imposition (Layout) Controls shall include the options enumerated in Table 3.1. *(L2-UI-234)*

Table 3.1.: *Boomaga-IPP GUI Application* Primary Layout Controls SETTINGS Options.

The *Boomaga-IPP GUI Application* shall present the user with following primary imposition options:[1]

1. 1 UP,
2. 2 UP,
3. 4 UP HORIZONTAL,
4. 4 UP VERTICAL,
5. 8 UP HORIZONTAL,
6. 8 UP VERTICAL,
7. BOOKLET, and
8. DOUBLE-SIDED.[2]

[1] The N-UP and BOOKLET options are all mutually exclusive.
[2] DOUBLE-SIDED is selected, but unavailable ("grayed-out") when BOOKLET is selected.

(L2-UI-234)

3.3.2. Stimulus/Response Sequences

The user shall command the desired file layout by selecting the desired imposition (layout) state through *UI* option buttons or equivalent functionality. *(L2-UI-068)*

3.3.3. Functional Requirements

The GUI shall not modify the contents of the JOB_STORE, but shall command updates to an output sequence state object which shall fully describe the desired output PDF file (print job) imposition (layout). *(L2-UI-069)* The output sequence state object shall be updated whenever the user changes the desired file imposition (layout) state through the GUI. *(L2-UI-070)*

3.4. Feature 4: Secondary Imposition (Layout) Controls

The Secondary Imposition (Layout) Controls shall be accessed through the CONFIGURE... button under the Downstream Printer Selection item [*(L2-UI-029)*, *(L2-UI-073)* & *(L2-UI-074)*], as they are in *Boomaga*. *(L2-UI-209)*

3.4.1. Description and Priority

When selected by the user, the CONFIGURE... button shall produce a popup menu window with a PROFILES subfeature and with SETTINGS (default) and MARGINS tabs from which the user may make further selections. *(L2-UI-210)* The *Boomaga* CONFIGURE... popup menu with the SETTINGS tab selected and displaying the *Boomaga* PROFILES subfeature is shown in Figure 3.2. The PROFILES subfeature shall allow the user to create, select, edit, and delete stored sets of SETTINGS and MARGINS for use at a later time. *(L2-UI-211)* The *Boomaga-IPP GUI Application* CONFIGURE... popup menu SETTINGS tab shall allow the user to select the imposition options enumerated in Table 3.2. *(L2-UI-212)*

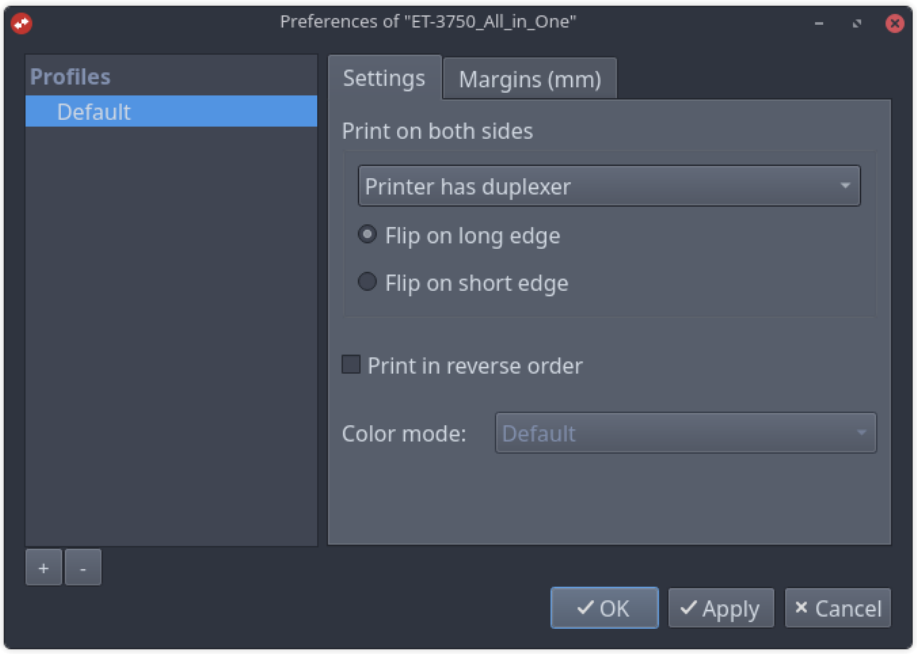

Figure 3.2.: *Boomaga* CONTINUE... Popup Menu with PROFILES Subfeature and SETTINGS Tab.

The *Boomaga* CONFIGURE... popup menu with the MARGINS tab selected and displaying the *Boomaga* PROFILES subfeature is shown in Figure 3.3. The *Boomaga-IPP GUI Application* CONFIGURE... popup menu MARGINS Tab shall allow the user to select the imposition options enumerated in Table 3.3. *(L2-UI-213)*

3.4.2. Stimulus/Response Sequences

Profiles Subfeature

The *Boomaga-IPP GUI Application* CONFIGURE... popup menu PROFILES Subfeature shall include a button to add a new profile. *(L2-UI-214)* Selecting the textscAdd Profile button shall immediately create a new profile, select it and allow the user to rename it or accept the system generated name. *(L2-UI-215)* The PROFILES Subfeature shall include a button to delete the selected profile. *(L2-UI-216)*

Table 3.2.: *Boomaga-IPP GUI Application* Secondary Layout Controls SETTINGS Options.

The *Boomaga-IPP GUI Application* shall present the user with following options on the SETTINGS tab of the CONFIGURE... popup menu:
1. Duplex printing using:
  a) The printer duplexer (if available) with:
    i. FLIP ON LONG EDGE or
    ii. FLIP ON SHORT EDGE,
  b) Manual duplexing with reverse printing,
  c) Manual duplexing without reverse printing,
2. PRINT IN REVERSE ORDER, and
3. COLOR MODE (AUTO/COLOR/GRAYSCALE).[1]

[1] Grayscale shall be applied by the IMPOSITION_ENGINE independently of the DOWNSTREAM_-PRINTER.

(L2-UI-212)

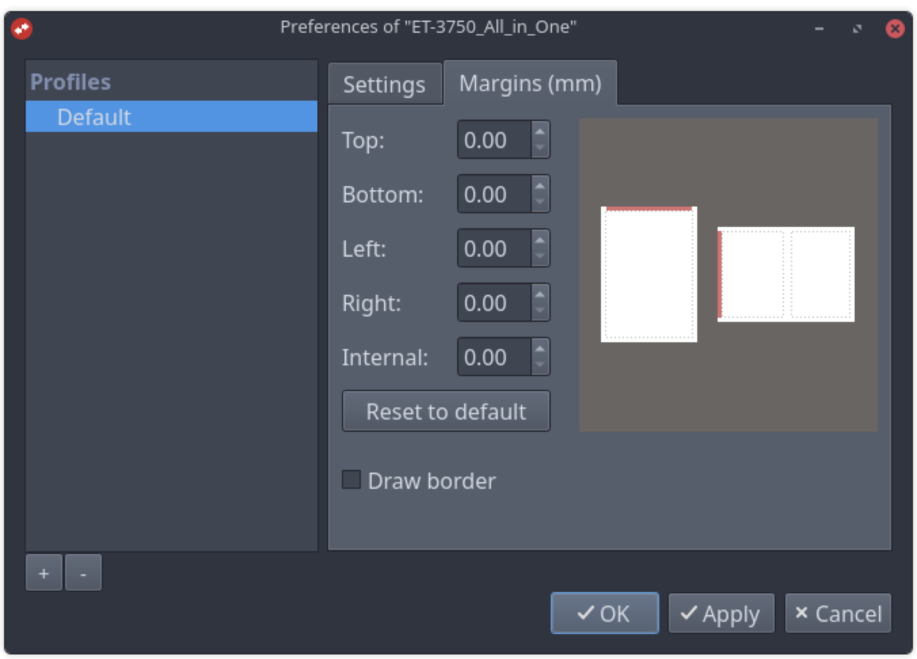

Figure 3.3.: *Boomaga* CONTINUE... Popup Menu with PROFILES Subfeature and MARGINS Tab.

Table 3.3.: *Boomaga-IPP GUI Application* Secondary Layout Controls MARGINS Options.

The *Boomaga-IPP GUI Application* shall present the user with following options on the MARGINS tab of the CONFIGURE... popup menu:

1. Control of the following printer spread margins in millimeters (mm):
   a) TOP,
   b) BOTTOM,
   c) LEFT,
   d) RIGHT, and
   e) INTERNAL (binding gutter);
2. Reset margin options to DEFAULT settings; and
3. DRAW BORDER around each imposed printer spread.

(L2-UI-213)

Settings Tab

The *Boomaga-IPP GUI Application* CONFIGURE... popup menu SETTINGS Tab shall include a menu allowing the user to select any of the three printer duplexing methods specified in *(L2-UI-212)*. *(L2-UI-217)* FLIP ON LONG EDGE and FLIP ON SHORT EDGE shall be selectable using mutually exclusive option buttons ("radio buttons") immediately below the duplexing option menu. *(L2-UI-218)* The FLIP ON LONG EDGE and FLIP ON SHORT EDGE "radio buttons" shall be available when use of the printer duplexer is selected and unavailable ("grayed out") whenever either manual duplexing menu option is selected. *(L2-UI-219)* PRINT IN REVERSE ORDER shall be selectable from the SETTINGS Tab via an independent, always-available option box. *(L2-UI-220)* DOWNSTREAM_PRINTER color mode shall be selectable from the SETTINGS Tab via a menu that shall be unavailable ("grayed out") if the currently selected DOWNSTREAM_PRINTER makes no color modes available. *(L2-UI-221)*

Margins Tab

The *Boomaga-IPP GUI Application* CONFIGURE... popup menu MARGINS Tab shall provide individual numeric entry boxes for the TOP, BOTTOM, LEFT, RIGHT, and INTERNAL (binding gutter) margins. *(L2-UI-222)* These numeric entry boxes shall be keyboard settable and shall accept values of two significant decimal places between 0.00 and 99.99. *(L2-UI-223)* When ALLOW NEGATIVE PAGE MARGINS *(L2-UI-173)* is enabled, these numeric entry boxes shall shall also accept values of two significant decimal places between 0.00 and -99.99. *(L2-UI-224)* The *Boomaga-IPP GUI Application* CONFIGURE... popup menu MARGINS Tab shall provide a RESET TO DEFAULT button that shall reset all five margins to their default values whenever selected by the user. *(L2-UI-225)* DRAW BORDER shall be selectable from the SETTINGS Tab via an independent, always-available option box. *(L2-UI-226)*

3.4.3. Functional Requirements

The *Boomaga-IPP GUI Application* CONFIGURE... popup menu PROFILES subfeature shall immediately save whatever SETTINGS and MARGINS values and options the user has set to the selected profile whenever the ✔ OK button or the ✔ Apply button are selected. *(L2-UI-227)* Selection of the

✔ OK button shall result in the *Boomaga-IPP* CONFIGURE... popup menu being closed as soon as the saving of any profile edits and of any changes to SETTINGS or MARGINS have been completed. *(L2-UI-228)* Selection of the ✔ Cancel button shall result in the *Boomaga-IPP* CONFIGURE... popup menu being closed immediately without saving any changes the user may have made to PROFILES, SETTINGS, or MARGINS. *(L2-UI-229)* Any saved changes to margin, gutter, and border made by the user through the *Boomaga-IPP GUI Application* shall be passed immediately to the IMPOSITION_ENGINE and shall be reflected immediately in the JOB_STORE state and in the JOB_STORE_PDF_PREVIEW. *(L2-UI-230)*

3.5. Feature 5: Downstream Printer Controls and Settings

3.5.1. Description and Priority

The *Boomaga-IPP* Downstream Printer Selection and Settings Controls shall be consolidated into a single contiguous region of the GUI screen immediately below the Menu Bar. *(L2-UI-071)*

3.5.2. Stimulus/Response Sequences

The Downstream Printer Selection and Settings controls shall emulate the standard menu structure of the *Linux*® system and its installed desktop environment as much as possible within the constraints of *Xilem*'s currently available widget functionality. *(L2-UI-072)*

3.5.3. Functional Requirements

The Downstream Printer Selection and Settings controls shall permit the user to select the desired Downstream Printer and permit the user to adjust its settings through *CUPS*-provided functionality. *(L2-UI-073)* These controls shall permit the user to select any suitable Downstream Printer available through *CUPS*. *(L2-UI-074)*

4. Handling of Erroneous Input

The *Boomaga-IPP* GUI shall prevent erroneous input by disallowing the selection of incompatible options by the user through the GUI. *(L2-UI-075)* When user selection of an option is disallowed, an error message shall be displayed to explain the incompatibility. *(L2-UI-076)*

4.1. Handling of Erroneous User Option Selections

When Allow negative page margins *(L2-UI-173)* is disabled, the GUI shall reject margin values less than zero and display an explanatory message. *(L2-UI-231)*

5. Menu Bar

The menu structure described in this chapter shall be a hierarchical menu of the type commonly incorporated into the window decoration of modern windowing systems. *(L2-UI-077)* The File, Edit Page, Edit Job, Preferences, and Help tabs shall be the top-level tabs in this menu structure and shall be constantly displayed on the menu bar in the order listed. *(L2-UI-078)* When any of these top level tabs are selected by the user using the mouse or from the keyboard, the menu of items available immediately below that tab in the menu tree structure shall be displayed. *(L2-UI-079)* The *Boomaga* menu bar is shown below in Figure 5.1. The *Boomaga* menus shown in this chapter are examples only and are not to be interpreted as prescribed requirements for *Boomaga-IPP* except as described explicitly in the text below.

Figure 5.1.: *Menu Bar.*

The items listed in the subsections below the top-level tab sections shall be user-selectable menu items. *(L2-UI-080)* Each menu item described in the subsections in this chapter shall initiate the action described in its subsection when selected by the user with the mouse or from the keyboard. *(L2-UI-081)*

When selected by the user, menu items with names followed by an ellipsis (...) shall produce a separate popup menu window from which the user may make further selections. *(L2-UI-082)* Menu item names followed by a right arrow (➔) shall produce a nested list within the menu bar menu tree structure from which the user may make selections while they are displayed. *(L2-UI-083)* Letters that are underlined in menu item names indicate that the keyboard combination alt + *underlined_key* shall open that particular menu item (e.g., alt + F from File opens the File Menu). *(L2-UI-084)* Keyboard shortcuts for menu items other than alt + *shortcut_key* are fully specified and enclosed in parentheses following the name of the menu item and shall open that menu item. *(L2-UI-085)*

5.1. File Menu

The File menu shall be opened by selecting the File tab from the menu bar with the mouse or by typing Alt+F from the keyboard. *(L2-UI-086)* The *Boomaga* File menu is shown below in Figure 5.2.

5.1.1. Print and Close (Ctrl+P)

Selecting this item from the menu or typing Ctrl+P from the keyboard shall immediately cause the imposition of the output print job as specified be the output sequence state object followed by the submission of the output print job to the downstream printer. *(L2-UI-087)* Upon successful submission, the *Boomaga-IPP* GUI shall clear the JOB_STORE. *(L2-UI-088)* Upon a successful clearing of the JOB_STORE, the *Boomaga-IPP* GUI shall quit. *(L2-UI-204)*

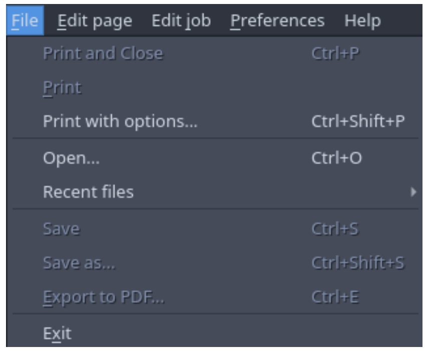

Figure 5.2.: *File Menu.*

5.1.2. Print

Selecting this item from the menu or typing Alt+P from the keyboard shall immediately cause the imposition of the output print job as specified be the output sequence state object followed by the submission of the output print job to the downstream printer. *(L2-UI-089)* The JOB_STORE and its state shall be unchanged by this action. *(L2-UI-090)*

5.1.3. Print with Options... (Ctrl+Shift+P)

Selecting this item from the menu or typing Ctrl+Shift+P from the keyboard shall immediately open the downstream printer options dialogue box (see Figure 5.3). *(L2-UI-091)* The print options dialogue box shall allow the user to select the number of copies of the output print job to be printed and to select collated or uncollated copies. *(L2-UI-092)* After selecting the desired printer options the user shall be able to select the ✔ OK button or the ✘ Cancel button. *(L2-UI-093)* Selecting the ✘ Cancel button shall close the printer options dialog, leaving the JOB_STORE state unchanged. *(L2-UI-094)* Selecting the ✔ OK button shall immediately modify and save the state of the JOB_STORE in accordance with the options selected by the user. *(L2-UI-095)* Upon successful update of the JOB_STORE state, the imposition of the output print job as specified be the output sequence state object shall be performed, immediately followed by the submission of the output print job to the downstream printer. *(L2-UI-096)* The JOB_STORE shall maintain its state at the conclusion of this action. *(L2-UI-097)*

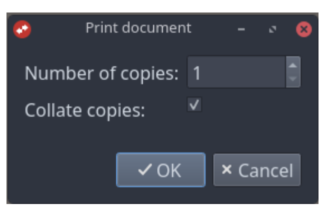

Figure 5.3.: *Printer Options Dialog Box.*

5.1.4. Open... (Ctrl+O)

Selecting this item from the menu or typing Ctrl+O from the keyboard shall immediately open the file selection dialogue box (see Figure 5.4). *(L2-UI-098)* The file selection dialogue box shall allow the user to select the ✕ Cancel button or to select any previously saved *Boomaga-IPP* JOB_STORE and state information file (.boo file) using the mouse. *(L2-UI-099)* Selecting the ✕ Cancel button shall close the file selection dialog, leaving the JOB_STORE state unchanged. *(L2-UI-100)* If the user has selected a .boo file, selecting the Open button or typing Alt+O from the keyboard shall cause the GUI to command the immediate loading of the selected file into the JOB_STORE. *(L2-UI-101)* Upon successful loading of the file into the JOB_STORE, the file selection dialogue box shall immediately close. *(L2-UI-102)*

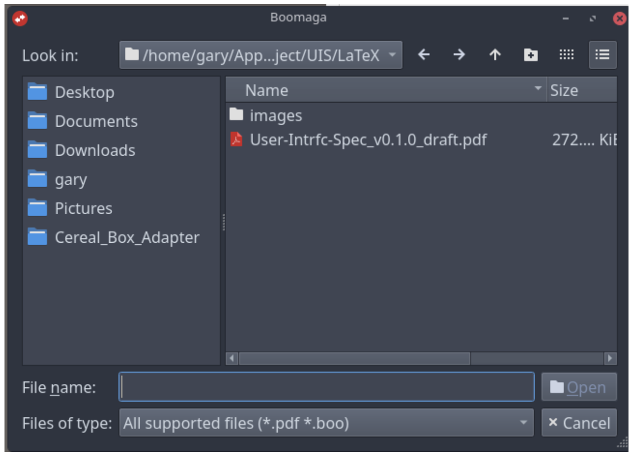

Figure 5.4.: *File Selection Dialogue Box.*

5.1.5. Recent Files →

If the user has not recently saved any .boo files, this menu item shall be greyed out and unavailable for selection. *(L2-UI-103)* If the user has recently saved any .boo files and the user selects this menu item, the it*Boomaga-IPP* UI shall display a pick list of recent .boo files. *(L2-UI-104)* If the user selects a file from the pick list, the UI shall command that the selected .boo file be immediately added to the current JOB_STORE. *(L2-UI-105)*

5.1.6. Save (Ctrl+S)

If the current JOB_STORE is null (empty) or if the user has not previously saved the current JOB_STORE using the Save As menu item, this item shall be grayed out and unavailable for selection. *(L2-UI-106)* If this item is available and the user selects it with the mouse or types Ctrl+S from the keyboard, the *Boomaga-IPP* UI shall command the immediate saving of the JOB_STORE and associated state information to the previously created .boo file, overwriting it. *(L2-UI-107)*

5.1.7. Save as... (Ctrl+Shift+S)

If the current JOB_STORE is null (empty), this item shall be grayed out and unavailable for selection. *(L2-UI-108)* Selecting this item from the menu or typing Ctrl+Shift+S from the keyboard shall immediately open the file save dialogue box (see Figure 5.5). *(L2-UI-109)* The file save dialogue box shall allow the user to select the ✕ Cancel button or to select a folder and type a filename for saving the *Boomaga-IPP* JOB_STORE and state information file (.boo file) using the mouse and keyboard. *(L2-UI-110)* Selecting the ✕ Cancel button shall close the file save dialog, leaving the JOB_STORE state unchanged. *(L2-UI-111)* If the user has selected a folder and typed a filename for the .boo file, selecting the Save button or typing Alt+S from the keyboard shall cause the *Boomaga-IPP* UI to command the immediate saving of the current JOB_STORE and associated state information into the desired file. *(L2-UI-112)* Upon successful saving of the JOB_STORE and state into the file, the file save dialogue box shall immediately close. *(L2-UI-113)*

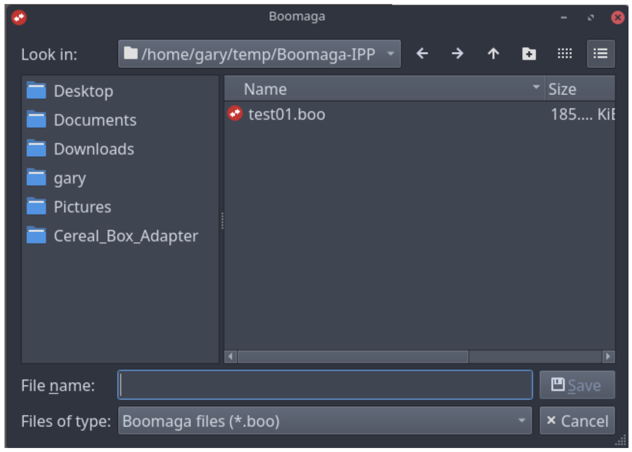

Figure 5.5.: *File Save Dialogue Box.*

5.1.8. Export to PDF... (Ctrl+E)

If the current JOB_STORE is null (empty), this item shall be grayed out and unavailable for selection. *(L2-UI-114)* If this menu item is available and selected by the user with the mouse or by typing Ctrl+E from the keyboard, the *Boomaga-IPP* UI shall immediately open the file export dialogue box (see Figure 5.6). *(L2-UI-115)* The file export dialogue box shall allow the user to select the ✕ Cancel button or to select a folder, type a filename, and enter metadata for saving the *Boomaga-IPP* output to a PDF file using the mouse and keyboard. *(L2-UI-116)* Selecting the ✕ Cancel button shall close the file export dialog, leaving the JOB_STORE state unchanged. *(L2-UI-117)* If the user has selected a folder and typed a filename for the PDF file, selecting the ✔ OK button shall cause the *Boomaga-IPP* UI to command the immediate saving of the imposed output of the current JOB_STORE into the desired file. *(L2-UI-118)* Upon successful saving of the output into the PDF file, the file export dialogue box shall immediately close. *(L2-UI-119)*

5.1.9. Exit

If this menu item is selected by the user with the mouse or by typing Alt+X from the keyboard, the *Boomaga-IPP* UI shall immediately save its current state information to the JOB_STORE temporary storage location. *(L2-UI-120)* Upon successful saving of the state information, it shall immediately exit. *(L2-UI-121)*

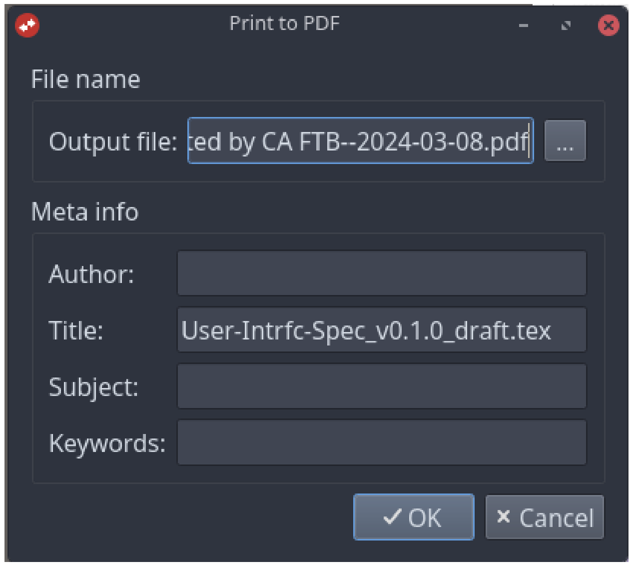

Figure 5.6.: *File Export Dialogue Box.*

5.2. Edit Page Menu

The Edit Page Menu shall be opened by selecting the Edit Page tab from the menu bar with the mouse or by typing Alt+E from the keyboard. *(L2-UI-122)* All Edit Page Menu commands shall act on, immediately preceding, or immediately following the currently selected document page or document pages, as appropriate to the selected command. *(L2-UI-201)* The *Boomaga* Edit page menu is shown below in Figure 5.7.

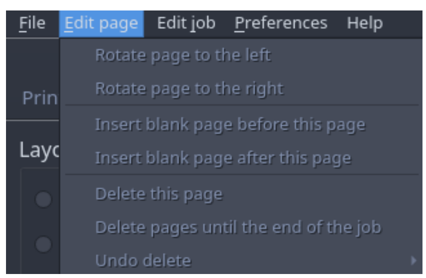

Figure 5.7.: *Edit Page Menu.*

5.2.1. Start new booklet from this page

This menu item shall be grayed out and unavailable for selection unless 1) the JOB_STORE is non-null and 2) the user has selected an output page from the JOB_STORE other than the first document page of the first job. *(L2-UI-123)* If the user has first selected a document page other the first document page of the first job from the JOB_STORE, and then selected this menu item with the mouse, the *Boomaga-IPP* UI shall immediately update the JOB_STORE current state information to reflect the creation of a new sub-booklet starting with the currently selected page. *(L2-UI-124)* The sub-booklets tab of the JOB_STORE Contents display shall then be updated to reflect the updated state of the JOB_STORE. *(L2-UI-125)*

5.2.2. Don't start new booklet from this page

This menu item shall be visible and available only when the currently selected document page is a manual sub-booklet start, otherwise the START NEW BOOKLET FROM THIS PAGE item shall be visible. *(L2-UI-238)* When the DON'T START NEW BOOKLET FROM THIS PAGE menu item is selected, the *Boomaga-IPP GUI Application* shall update the JOB_STORE current state to remove the manual sub-booklet break at that page, and the Sub-booklets Tab of the JOB_STORE_CONTENTS_LISTING display shall be updated accordingly. *(L2-UI-239)*

5.2.3. Rotate page counterclockwise

This menu item shall be grayed out and unavailable for selection unless the JOB_STORE is non-null. *(L2-UI-126)* If the user has selected this menu item with the mouse, the *Boomaga-IPP* UI shall immediately update the JOB_STORE current state information to reflect the 90 degree counterclockwise rotation of the currently selected document page from its previous state. *(L2-UI-127)* The JOB_STORE_-PDF_PREVIEW shall then be updated to reflect the updated state of the the selected document page of the JOB_STORE. *(L2-UI-128)*

5.2.4. Rotate page clockwise

This menu item shall be grayed out and unavailable for selection unless the JOB_STORE is non-null. *(L2-UI-129)* If the user has selected this menu item with the mouse, the *Boomaga-IPP* UI shall immediately update the JOB_STORE current state information to reflect the 90 degree clockwise rotation of the currently selected document page from its previous state. *(L2-UI-130)* The JOB_STORE_PDF_PREVIEW shall then be updated to reflect the updated state of the the selected document page of the JOB_STORE. *(L2-UI-131)*

5.2.5. Insert blank page before this page

This menu item shall be grayed out and unavailable for selection unless the JOB_STORE is non-null. *(L2-UI-132)* If the user has first selected this menu item with the mouse, the *Boomaga-IPP* UI shall immediately update the JOB_STORE current state information to reflect the addition of a blank page immediately preceding the currently selected document page. *(L2-UI-133)* The JOB_STORE_PDF_PREVIEW shall then be updated to reflect the addition of the blank page to the JOB_STORE. *(L2-UI-134)*

5.2.6. Insert blank page after this page

This menu item shall be grayed out and unavailable for selection unless the JOB_STORE is non-null. *(L2-UI-135)* If the user has selected this menu item with the mouse, the *Boomaga-IPP* UI shall immediately update the JOB_STORE current state information to reflect the addition of a blank page immediately following the currently selected document page. *(L2-UI-136)* The JOB_STORE_PDF_PREVIEW shall then be updated to reflect the addition of the blank page to the JOB_STORE. *(L2-UI-137)*

5.2.7. Delete this page

This menu item shall be grayed out and unavailable for selection unless the JOB_STORE is non-null. *(L2-UI-138)* If the user has selected this menu item with the mouse, the *Boomaga-IPP* UI shall immediately update the JOB_STORE current state information to reflect the deletion of the previously selected document page from the JOB_STORE. *(L2-UI-139)* The JOB_STORE_PDF_PREVIEW shall then be updated to reflect the deletion of the document page from the JOB_STORE. *(L2-UI-140)* The document page selected shall be updated to the document page preceding the deleted document page, or to the first document page if there was no preceding document page. *(L2-UI-141)*

5.2.8. Delete pages to the end of the job

This menu item shall be grayed out and unavailable for selection unless the JOB_STORE is non-null. *(L2-UI-142)* If the user has selected this menu item with the mouse, the *Boomaga-IPP* UI shall immediately update the JOB_STORE current state information to reflect the deletion of all document pages in the JOB_STORE from the previously selected document page through the end of the current job. *(L2-UI-143)* The JOB_STORE_PDF_PREVIEW shall then be updated to reflect the deletion of these pages from the JOB_STORE. *(L2-UI-144)* The document page selected shall be updated to the document page preceding the first deleted document page. *(L2-UI-145)*

5.2.9. Undo delete →

This menu item shall be grayed out and unavailable for selection unless the JOB_STORE is non-null and the user has previously deleted one or more document pages. *(L2-UI-146)* If the user has selected this menu item with the mouse, the *Boomaga-IPP* UI shall immediately list the the previously deleted JOB_STORE document pages in the order of their original appearance in the JOB_STORE. *(L2-UI-147)* If the user then selects one of the previously deleted document pages from the list with either the mouse or the keyboard arrow and carriage return keys, the JOB_STORE state shall immediately be updated to show that page as restored to its original position in the JOB_STORE. *(L2-UI-148)* The JOB_STORE_PDF_PREVIEW shall then be updated to reflect the undeletion of the document page from the JOB_STORE. *(L2-UI-149)* The undelete pages list shall also be updated to remove the undeleted document page from the list. *(L2-UI-150)*

5.3. Edit Job Menu

The Edit job menu shall be opened by selecting the Edit job tab from the menu bar with the mouse or by typing Alt+J from the keyboard. *(L2-UI-151)* The *Boomaga* Edit job menu is shown below in Figure 5.8.

5.3.1. Rename Job

This menu item shall be grayed out and unavailable for selection unless the JOB_STORE is non-null. *(L2-UI-152)* If the JOB_STORE is non-null and the user has selected this menu item with the mouse, the *Boomaga-IPP* GUI shall immediately display the Rename Job menu for the selected print job, allowing the user to enter a new name for that print job. *(L2-UI-153)* Upon the user entering a new name and

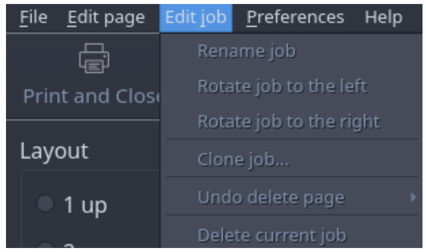

Figure 5.8.: *Edit Job Menu.*

then selecting the ✔ OK button, the JOB_STORE state shall be immediately updated with the user provided name and the JOB_STORE_CONTENTS_LISTING shall then be updated to reflect the renaming of the selected print job. *(L2-UI-154)* The *Rename Job* menu is shown below in Figure 5.9.

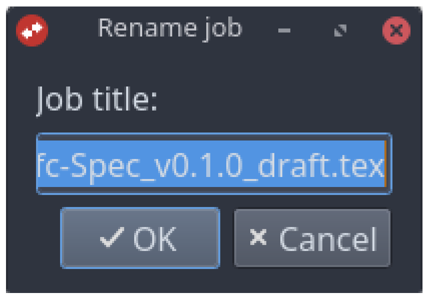

Figure 5.9.: *Rename Job Menu.*

5.3.2. Rotate job counterclockwise

This menu item shall be grayed out and unavailable for selection unless the JOB_STORE is non-null. *(L2-UI-155)* If the JOB_STORE is non-null and the user has selected this menu item with the mouse, the *Boomaga-IPP* GUI shall immediately update the JOB_STORE state for all document pages in the selected print job to reflect a 90 degree counterclockwise rotation from the previous orientation of each affected document page. *(L2-UI-156)* The JOB_STORE_PDF_PREVIEW shall then be updated to display the revised rotation of the document pages of the selected print job. *(L2-UI-157)*

5.3.3. Rotate job clockwise

This menu item shall be grayed out and unavailable for selection unless the JOB_STORE is non-null. *(L2-UI-158)* If the JOB_STORE is non-null and the user has selected this menu item with the mouse, the *Boomaga-IPP* GUI shall immediately update the JOB_STORE state for all document pages in the selected print job to reflect a 90 degree clockwise rotation from the previous orientation of each affected document page. *(L2-UI-159)* The JOB_STORE_PDF_PREVIEW shall then be updated to display the revised rotation of the document pages of the selected print job. *(L2-UI-160)*

5.3.4. Clone job...

This menu item shall be grayed out and unavailable for selection unless the JOB_STORE is non-null. *(L2-UI-161)* If the JOB_STORE is non-null and the user has selected this menu item with the mouse, the *Boomaga-IPP* GUI shall immediately command the cloning of a copy of the selected print job onto the end of the JOB_STORE. *(L2-UI-162)* The JOB_STORE_CONTENTS_LISTING and the JOB_STORE_PDF_PREVIEW shall then be updated to display the newly cloned print job following the print job that was previously last in the JOB_STORE. *(L2-UI-163)*

5.3.5. Undo delete page →

This menu item shall be grayed out and unavailable for selection unless 1) the JOB_STORE is non-null and 2) the user has previously deleted an output page from the selected print job of the JOB_STORE. *(L2-UI-164)* If the user has first deleted a page from the selected print job, and then selected this menu item with the mouse, the *Boomaga-IPP* GUI shall immediately list the the previously deleted document pages in the order of their original appearance in the JOB_STORE. *(L2-UI-165)* If the user then selects one of the previously deleted document pages from the list with either the mouse or the keyboard arrow and carriage return keys, the JOB_STORE state shall immediately be updated to show that document page as restored to its original position in the JOB_STORE. *(L2-UI-166)* The JOB_STORE_PDF_PREVIEW shall then be updated to reflect the restoration of the page to the JOB_STORE. *(L2-UI-167)* The undelete pages list shall also be updated to remove the undeleted document page from the list. *(L2-UI-168)*

5.3.6. Delete current job

This menu item shall be grayed out and unavailable for selection unless the JOB_STORE is non-null. *(L2-UI-169)* If the JOB_STORE is non-null and the user has selected this menu item with the mouse, the *Boomaga-IPP* GUI shall immediately command the deletion of the selected print job from the JOB_STORE. *(L2-UI-170)* The JOB_STORE_CONTENTS_LISTING and the JOB_STORE_PDF_PREVIEW shall then be updated to reflect the removal of the print job from the JOB_STORE. *(L2-UI-171)*

5.4. Preferences Menu

The Preferences menu shall be opened by selecting the Preferences tab from the menu bar with the mouse or by typing Alt+P from the keyboard as shown in Figure 5.10. *(L2-UI-172)*

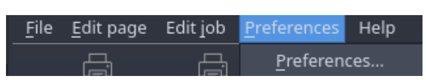

Figure 5.10.: *Preferences Tab.*

5.4.1. Preferences...

From the Preferences Menu, the user shall be able to select the following option boxes: 1) Print as sub-booklets, 2) AutoSave the recent sessions, 3) Right-to-left direction, and 4) Allow negative page margins. *(L2-UI-173)* If the user selects "Print as sub-booklets," then they shall be able to set the number of "Sheets per sub-booklet."*(L2-UI-174)* The default value of "Sheets per sub-booklet" shall be 20. *(L2-UI-175)* If the user selects "AutoSave the recent sessions," then they shall be able to select the directory in which recent session files are saved using the display environment's standard means of selecting a directory through its GUI. *(L2-UI-176)* The *Boomaga* Preferences menu is shown below in Figure 5.11.

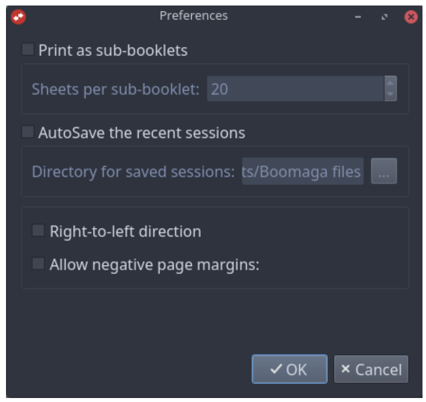

Figure 5.11.: *Preferences Menu.*

5.5. Help Menu

The Help menu shall be opened by selecting the Help tab from the menu bar with the mouse as shown in Figure 5.12. *(L2-UI-177)*

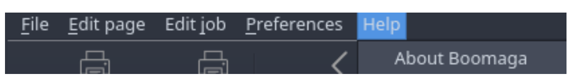

Figure 5.12.: *Help Menu.*

5.5.1. About Boomaga-IPP

The About *Boomaga-IPP* menu window shall be opened by selecting "About *Boomaga-IPP*" from the "Help" menu. *(L2-UI-178)* The window shall provide information on *Boomaga-IPP* including version number, copyright information, project software location, software license information, a list of authors, a list of acknowledgements, information about available translations, and any other TBD information (TBD-UIS-002)[1] that the *Boomaga-IPP* team should decide to include. *(L2-UI-179)* The About *Boomaga* pop-up window is shown as an example in Figure 5.13.

[1]See Appendix D, To Be Determined (TBD) List, for all current UIS TBDs.

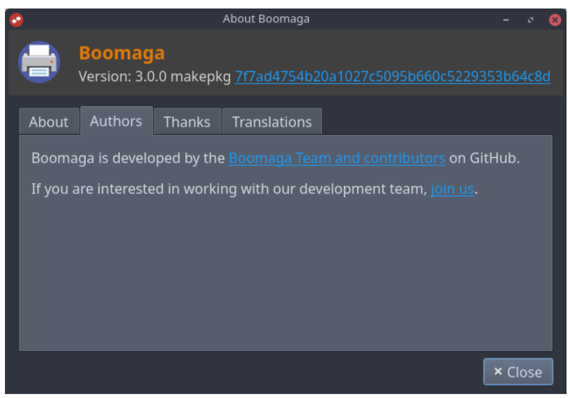

Figure 5.13.: *About Boomaga Menu.*

6. Context Specific Menus

6.1. Context Specific Edit Page Menu

The user shall be able to open a context specific Edit page menu with a "right click" of the mouse over the page displayed in the JOB_STORE_PDF_PREVIEW Feature. *(L2-UI-180)* The context specific Edit page menu and its items shall be identical in function to the those of the menu bar Edit page menu described in Section 5.2. *(L2-UI-181)* An example context specific Edit page menu is shown in Figure 6.1.

6.2. Context Specific Edit Job Menu

The user shall be able to open a context specific Edit job menu with a "right click" of the mouse over a print job displayed in the JOB_STORE_CONTENTS_LISTING Feature. *(L2-UI-182)* The context specific Edit job menu and its items shall be identical in function to the those of the menu bar Edit job menu described in Section 5.3. *(L2-UI-183)* An example context specific Edit job menu is shown in Figure 6.2.

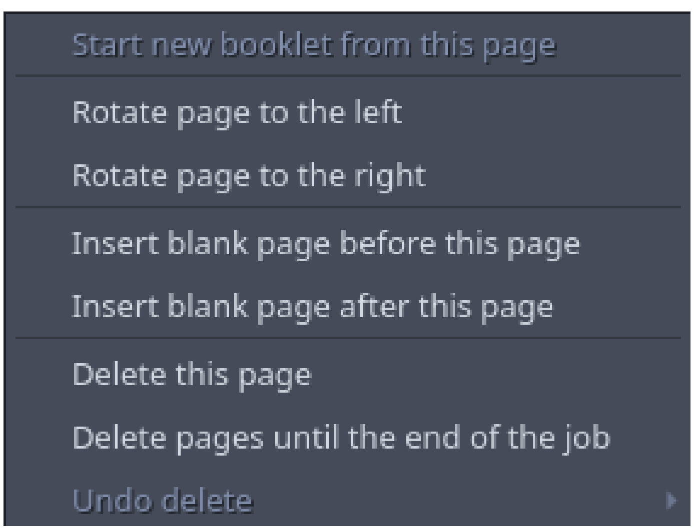

Figure 6.1.: *Context Specific Edit Page Menu.*

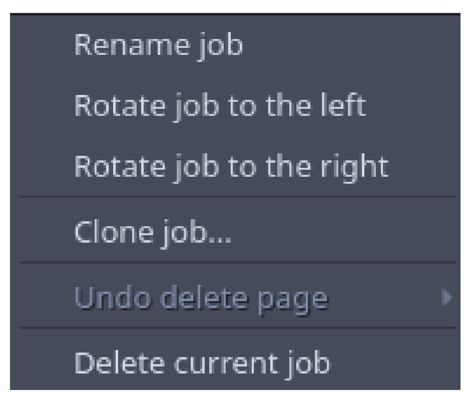

Figure 6.2.: *Context Specific Edit Page Menu.*

7. Miscellaneous Interface Elements

7.1. Application Window Decoration

The application window decoration shall display the name of the program, *Boomaga-IPP*, centered on the top border of the application window. *(L2-UI-184)* The application window decoration shall display the following information right justified in the bottom border of the application window: 1) The number of the imposed page currently displayed in JOB_STORE_PDF_PREVIEW Feature out of the total number of imposed pages in the JOB_STORE, 2) The number of the document page currently selected out of the total number of document pages in the JOB_STORE, and 3) The total number of sheets to be printed in the JOB_STORE. *(L2-UI-185)* The above information shall be updated as required whenever the user makes a change to the imposition controls or adds or removes a document page. *(L2-UI-186)* An example of this information as displayed by *Boomaga* is shown in Figure 7.1.

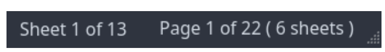

Figure 7.1.: *Page Number Information in Boomaga Bottom-Right Window Frame.*

8. Integration with the Operating System

8.1. *IPP Everywhere™* Integration

The *Boomaga-IPP IPP Everywhere™* client shall launch the *Boomaga-IPP* GUI application on the user's desktop whenever the user prints to *Boomaga-IPP*. *(L2-UI-187)*

8.2. Job_Store Interaction

The *Boomaga-IPP* GUI shall not manipulate the Job_Store directly, but shall communicate with the *Boomaga-IPP* Imposition_Engine, which shall manipulate the Job_Store as directed by the user via the GUI. *(L2-UI-188)* The *Boomaga-IPP* GUI shall communicate user-desired changes to the state of the Job_Store and access the updated Job_Store state in order to display the correct and up-to-date Job_Store_PDF_Preview and Job_Store_Contents_Listing to the user. *(L2-UI-189)*

8.3. *CUPS* Integration

The *Boomaga-IPP* GUI application shall communicate with the *Boomaga-IPP* Downstream_-Printer to print the imposed Job_Store to the user-selected downstream printer. *(L2-UI-190)* If the option is selected by the user, the *Boomaga-IPP* GUI shall request that the Job_Store be flushed and the *Boomaga-IPP* GUI application quit upon successful printing of the imposed Job_Store. *(L2-UI-191)*

9. Special Considerations

9.1. Translation Requirements

Boomaga-IPP shall be designed to allow the GUI to display text in languages other than English. *(L2-UI-192)*

9.2. Reuse Requirements

The *Boomaga-IPP Virtual Printer* system shall use existing well documented and well maintained software libraries whenever possible so as ease development difficulty and improve reliability. *(L2-UI-193)* These libraries shall include non-*RUST®* libraries with safe *RUST®* bindings, such as *Poppler*, *qpdf*, and *D-Bus*. *(L2-UI-194)* The *Boomaga-IPP* JOB_STORE_PDF_PREVIEW feature shall be designed to support its reuse in a stand-alone PDF file viewer. *(L2-UI-202)* The *Boomaga-IPP* CAPTURE_DAEMON shall be designed to support its reuse in other printer or virtual printer applications. *(L2-UI-203)*

Appendix A.

Glossary

Ada A modern programming language designed for large, long-lived applications – and embedded systems in particular – where reliability and efficiency are essential. It was most recently updated in 2022. Those revisions, know as Ada 2022, have been published as the International Standard ISO/IEC 8652:2023(E). For additional information, see Section 1.8. References: ISO/IEC JTC 1/SC 22/WG 9 - Ada, 2023.

Adobe® A registered trademark of Adobe, Inc..

ARCH LINUX™ *A simple, lightweight Linux® distribution™*. For additional information, see Section 1.8. References: Vinet et al., nodate.

Boomaga-IPP GUI Application The application software that provides the GUI between the user and the *Boomaga-IPP Virtual Printer*. For additional information, see the *Boomaga-IPP Project Repository*: Section 1.8. References: Martin and Boomaga-IPP Project Team, 2026-03-03/2026a.

Boomaga An orphaned virtual *CUPS* printer system for *Linux®* computers. The last release, v3.0.0, was issued Mar 13, 2019. A patch to the v3.0.0 source code was merged into the *Boomaga* source code on February 20, 2022 with no subsequent release. *Boomaga* has been unmaintained since that date. For additional information, see Section 1.8. References: Sokolov et al., 2013b.

Boomaga-IPP An *IPP Everywhere™* / *CUPS* virtual printer development project for *Linux®* computers. For additional information, see the *Boomaga-IPP Project Repository*: Section 1.8. References: Martin and Boomaga-IPP Project Team, 2026-03-03/2026a.

Boomaga-IPP Project Repository The *Boomaga-IPP* Project Repository for software and documentation, currently a public repository on *GitHub®*. For additional information, see the *Boomaga-IPP* Project Respoitory: Section 1.8. References: Martin and Boomaga-IPP Project Team, 2026-03-03/2026a.

Boomaga-IPP Virtual Printer An *IPP Everywhere™* / *CUPS* virtual printer software system for *Linux®* computers using *Wayland* display environments and managed via *systemd*; written entirely in *RUST®*. For additional information, see the *Boomaga-IPP Project Repository*: Section 1.8. References: Martin and Boomaga-IPP Project Team, 2026-03-03/2026a.

Capture_Daemon The *Boomaga-IPP* client/server software component responsible for receiving (capturing) input print jobs as PDF files through an *IPP Everywhere™* print service and adding them to the JOB_STORE..

Cargo® The official package manager and build system for the *RUST®* programming language. For additional information, see Section 1.8. References: Rust Foundation, nodate-a.

CLAUDE® A series of LLMs developed by Anthropic PBC and first released in 2023. *CLAUDE®* is used for software development via *CLAUDE® Code*. For additional information, see Section 1.8. References: Anthropic PBC, nodate-a.

CLAUDE® Code An agentic AI coding tool developed by Anthropic PBC that reads entire codebases, edits files, runs commands, and manages git workflows through natural language prompts. It operates autonomously across multiple files to execute tasks such as building features, fixing bugs, and refactoring, while requiring explicit user permission for file modifications and command execution. For additional information, see Section 1.8. References: Anthropic PBC, nodate-b.

crates.io The *RUST®* community's shared software crate registry. For additional information, see Section 1.8. References: Rust Foundation, nodate-b.

CUPS A modular printing system for Unix-like computer operating systems which allows a computer to act as a print server. Formerly an acronym for Common UNIX Printing System. For additional information, see Section 1.8. References: Sweet and OpenPrinting, nodate.

D-Bus A desktop message bus system, a simple way for applications to talk to one another. *D-Bus* provides interprocess communication and helps coordinate process lifecycle; it makes it simple and reliable to code a "single instance" application or daemon, and to launch applications and daemons on demand when their services are needed. For additional information, see Section 1.8. References: Pennington et al., nodate.

document page A single page of a source PDF document (an input print job) captured by the *Boomaga-IPP* CAPTURE_DAEMON. Equivalent to a logical page.

Downstream_Printer The downstream physical or virtual printer to which the *Boomaga-IPP Virtual Printer* submits its imposed JOB_STORE through the host computer's *CUPS* server.

Druid A data-first, *RUST®*-native *UI* design toolkit designed to offer a polished user experience for cross-platform desktop applications. While the project is no longer under active development by the *Linebender Organization*, version 0.8.3 remains available on *crates.io* and on *GitHub®* for bug fixes and documentation improvements. For additional information, see Section 1.8. References: Levien et al., 2023.

GhostScript® A suite of software based on an interpreter for *Adobe®*'s *POSTSCRIPT®* and Portable Document Format (PDF) page description languages. Its main purposes are the rasterization of documents in these languages, the display or printing of document pages, and conversion between *POSTSCRIPT®* and PDF files. For additional information, see Section 1.8. References: Artifex Software, Inc., nodate.

git A free and open source distributed version control system designed to handle everything from small to very large projects with speed and efficiency. For additional information, see Section 1.8. References: Torvalds et al., nodate.

GitHub® A proprietary developer platform that allows developers to create, store, manage, and share their code. It uses git to provide distributed version control and *GitHub* itself provides access control, bug tracking, software feature requests, task management, continuous integration, and wikis for every project. For additional information, see Section 1.8. References: GitHub, Inc. (A subsidiary of Microsoft Corporation), 2026.

imposed Arranged onto a set of facing pages (a printer spread) in a specified orientation and order so as to produce a desired output from a printer.

imposed page A single page of a PDF output document (a print job) arranged and oriented as specified by the *Boomaga-IPP* user and ready to be submitted to the *Boomaga-IPP* Downstream_Printer. A set of facing imposed pages is referred to as a printer spread.

imposition The process of arranging and orienting (imposing) document pages onto printer spreads for printing.

Imposition_Engine The *Boomaga-IPP* client/server software component responsible for managing the Job_Store and its associated state information. The Imposition_Engine serves the imposed Job_Store to the *Boomaga-IPP GUI Application* Job_Store_PDF_Preview feature and to the *Boomaga-IPP* Downstream_Printer on demand.

IPP Everywhere™ An extension of IPP to support network printing without vendor-specific driver software, including the transport, various discovery protocols, and standard document formats. For additional information, see Section 1.8. References: Printer Working Group, 2020.

Job_Store The *Boomaga-IPP* software component containing the Job_Store data structure and related access methods. Print jobs received through the *IPP Everywhere™* client are added to the Job_Store. The *Boomaga-IPP* GUI accesses and modifies the user-commanded output state of the Job_Store. The *Boomaga-IPP* Downstream_Printer requests the imposed Job_Store and relays it to the user-selected downstream printer on request.

Job_Store_Contents_Listing The Job_Store_Contents_Listing feature of the *Boomaga-IPP* GUI (*UI* Feature 1).

Job_Store_PDF_Preview The Job_Store_PDF_Preview feature of the *Boomaga-IPP GUI Application* (*UI* Feature 2).

KDE An international community developing high-quality, free, and open-source software. KDE was originally an acronym for the K Desktop Environment, a wordplay on the pre-existing *Common Desktop Environment*. For additional information, see Section 1.8. References: KDE Community, 2026.

L2 Designates a Level 2 requirement within a requirement ID of the form LX-SS-nnn. LX would be L1 for Level 1 Requirements, L2 for Level 2, etc. SS is the subsystem: SYS for System Wide, UI for User Interface, etc.

Linebender Organization “A friendly group of people who share an interest in 2D graphics and user interface design, with everything that entails.” Most *Linebender* projects are developed in the Rust programming language. For additional information, see Section 1.8. References: Linebender Organization, nodate-a.

Linux® A very large family of *UNIX®*-like operating systems based on a kernel first produced by Linus Torvalds in 1991. For additional information, see Section 1.8. References: Linus Torvalds et al., nodate and Linux Foundation, nodate.

LLAMA® A family of AI models produced by *Meta AI*, a division of *Meta Platforms, Inc*. For additional information, see Section 1.8. References: Meta AI, nodate.

logical page A single page of a source PDF document (an input print job) captured by the *Boomaga-IPP* CAPTURE_DAEMON. Equivalent to a document page.

opencode® An open source AI coding agent produced by Anomaly Innovations Inc. For additional information, see Section 1.8. References: Anomaly Innovations Inc., nodate.

page One side of a sheet of paper. Within this document, logical page or document page refers to a single page of a source PDF document (a print job) captured by *Boomaga-IPP* CAPTURE_DAEMON. Imposed page refers to a single page imposed for submission to the *Boomaga-IPP* DOWNSTREAM_PRINTER.

PDF A file format developed by Adobe in 1993 and standardized as ISO 32000 in 2008. The PDF format is used to present documents, including text formatting and images, in a manner independent of application software, hardware, and operating systems. Based on the PostScript language, a PDF file encapsulates a complete description of a fixed-layout document, including the text, fonts, vector graphics, raster images and other information needed to display it. For additional information, see 1.8. References: ISO/TC 171/SC 2, 2008.

Perplexity.ai A worldwide web based tool for querying various online AI models, primarily its own search engine, known as Sonar, based on a Meta *LLAMA®* model. A product of *Perplexity AI, Inc*. For additional information, see Section 1.8. References: Srinivas et al., nodate.

PlantUML An open-source tool allowing users to create diagrams from a plain text language. Besides various UML diagrams, PlantUML has support for various other software development related formats (such as Archimate, Block diagram, BPMN, C4, Computer network diagram, ERD, Gantt chart, Mind map, and WBD), as well as visualisation of JSON and YAML files. For additional information, see Section 1.8. References: Roques and PlantUML Contributors, 2026.

Plasma KDE’s desktop environment. For additional information, see Section 1.8. References: KDE Community, nodate.

Poppler a PDF rendering library based on the xpdf-3.0 code base. For additional information, see Section 1.8. References: Høgsberg and The Poppler Project, nodate.

Poppler-rs A high-level (safe) set of Rust bindings for *Poppler*'s glib interface. *Poppler* is a PDF rendering library with a *cairo* backend. For additional information, see Section 1.8. References: Dröge et al., 2026.

POSTSCRIPT® A page description language and dynamically typed, stack-based programming language. For additional information, see Section 1.8. References: Adobe, Inc., nodate.

printer spread An imposed spread ready for submission to a printer.

qpdf A command-line tool and C++ library that performs content-preserving transformations on PDF files. It supports linearization, encryption, and numerous other features. It can also be used for splitting and merging files, creating PDF files (but you have to supply all the content yourself), and inspecting files for study or analysis. For additional information, see Section 1.8. References: Berkenbilt and Holger, nodate.

RUST® A general-purpose programming language noted for its emphasis on performance, type safety, concurrency, and memory safety. For additional information, see Section 1.8. References: Rust Foundation, nodate-c.

spread A set of facing pages in a book or booklet..

systemd A software suite for system and service management on *Linux®* built to unify service configuration and behavior across *Linux®* distributions. For additional information, see Section 1.8. References: Poettering and Sievers, nodate.

UI A subsystem identifier in the *Boomaga-IPP* requirements numbering scheme; used to identify user interface requirements. Also an acronym for user interface.

UNIX® A family of multi-tasking, multi-user computer operating systems that derive from the original AT&T Unix, the development of which started in 1969 at the Bell Labs research center by Ken Thompson, Dennis Ritchie, and others. For additional information, see Section 1.8. References: The Open Group, nodate.

Wayland A replacement for the X11 window system protocol and architecture with the aim to be easier to develop, extend, and maintain. *Wayland* is the language (protocol) that applications can use to talk to a display server in order to make themselves visible and get input from the user (a person). A *Wayland* server is called a "compositor." Applications are *Wayland* clients. For additional information, see see Section 1.8. References: Høgsberg and The Wayland Project, nodate.

X11 A windowing system for bitmap displays, common on Unix-like operating systems. X originated as part of Project Athena at Massachusetts Institute of Technology (MIT) in 1984. The X protocol has been at version 11 (hence "X11") since September 1987. For additional information, see Section 1.8. References: Gettys et al., nodate.

Xilem An experimental, high-performance *RUST® GUI Framework* under development by the Linebender community, designed to create cross-platform desktop and mobile applications. Inspired by *SwiftUI*, *Flutter*, and *Elm*, it utilizes a reactive architecture. For additional information, see Section 1.8. References: Linebender Organization, 2022-11-16/2026 and Linebender Organization, nodate-b.

zbus_systemd A *RUST®* library for communication with *systemd* processes. For additional information, see Section 1.8. References: Bruno and Khan, 2026.

Appendix B.

Acronyms

AI artificial intelligence.

FAQ frequently asked questions.

GUI graphical user interface.

IEEE Institute of Electrical and Electronic Engineers.

IPC inter-process communication.

IPP Internet Printing Protocol.

ISTO Industry Standards and Technology Organization.

LLM large language model.

PDF Portable Document Format.

PWG Printer Working Group.

SRS software requirements specification.

TBD to be determined.

UI user interface.

UIS user interface specification.

UML Unified Modeling Language.

Appendix C.
Analysis Models

Perplexity.ai was used to produce the following UML models based on the synthesis of the *Boomaga* virtual printer as currently implemented and the initial *Boomaga-IPP* top level requirements. *PlantUML* was then used to generate UML diagrams from the models.

Appendix C.1. Component Diagram

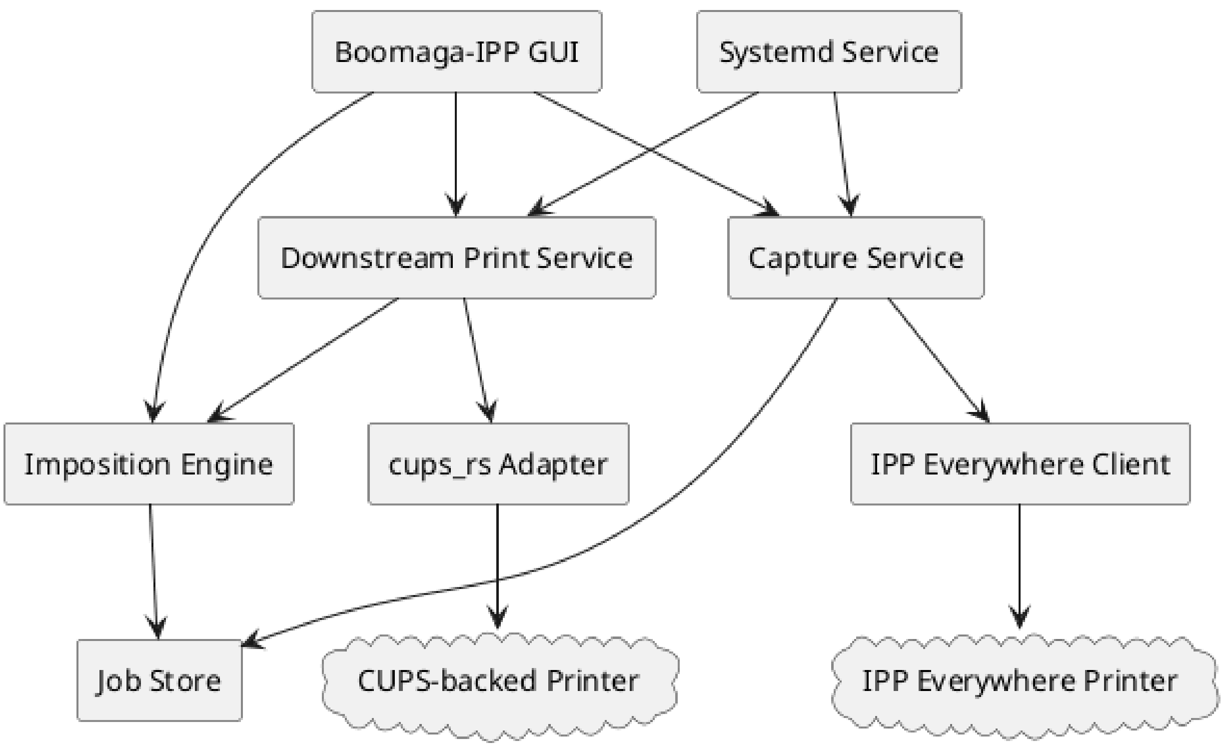

Figure C.1.: *Boomaga-IPP Virtual Printer* System Component Diagram.

Appendix C.2. Class Diagram

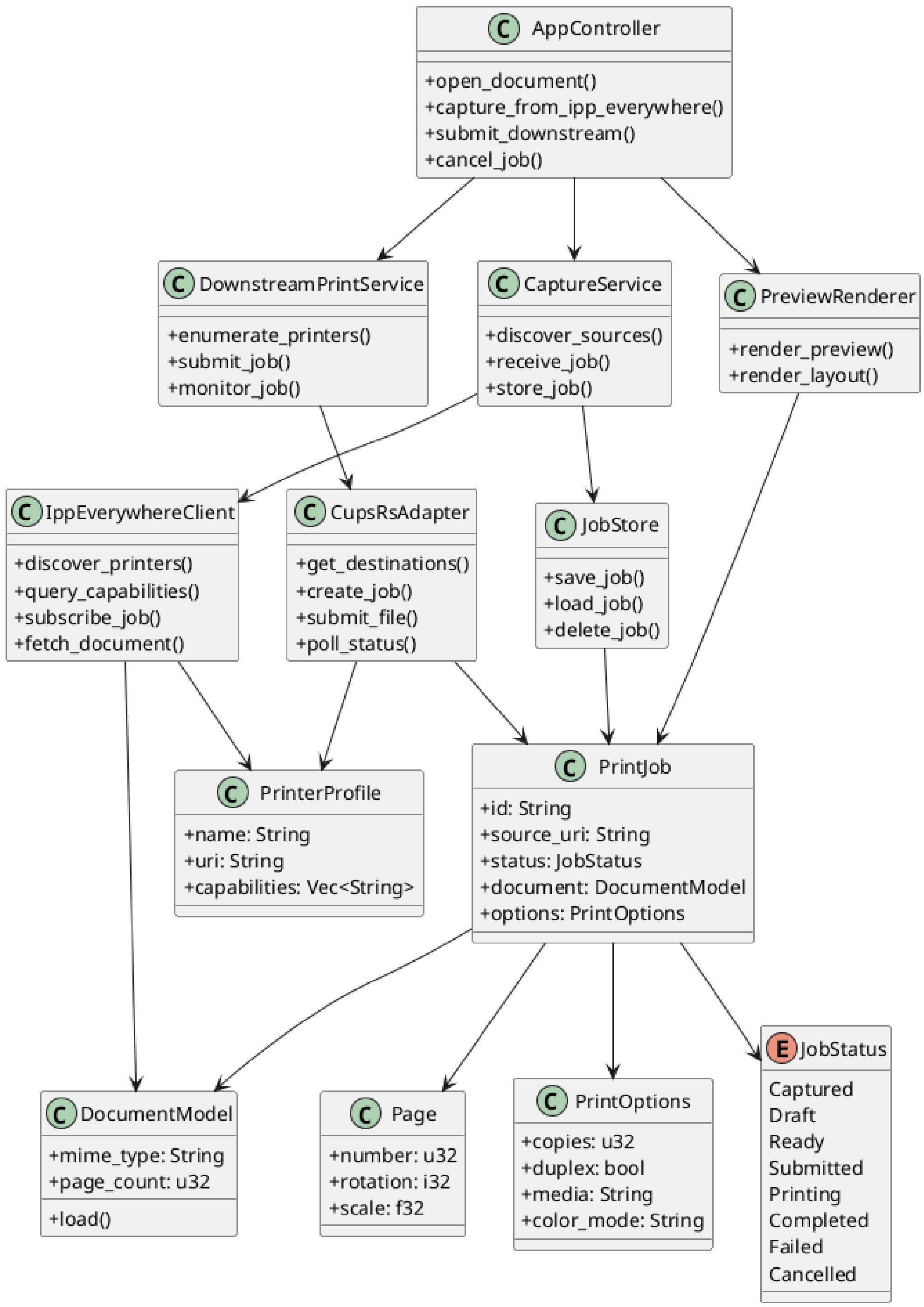

Figure C.2.: *Boomaga-IPP Virtual Printer* System Class Diagram.

Appendix C.3. Sequence Diagram

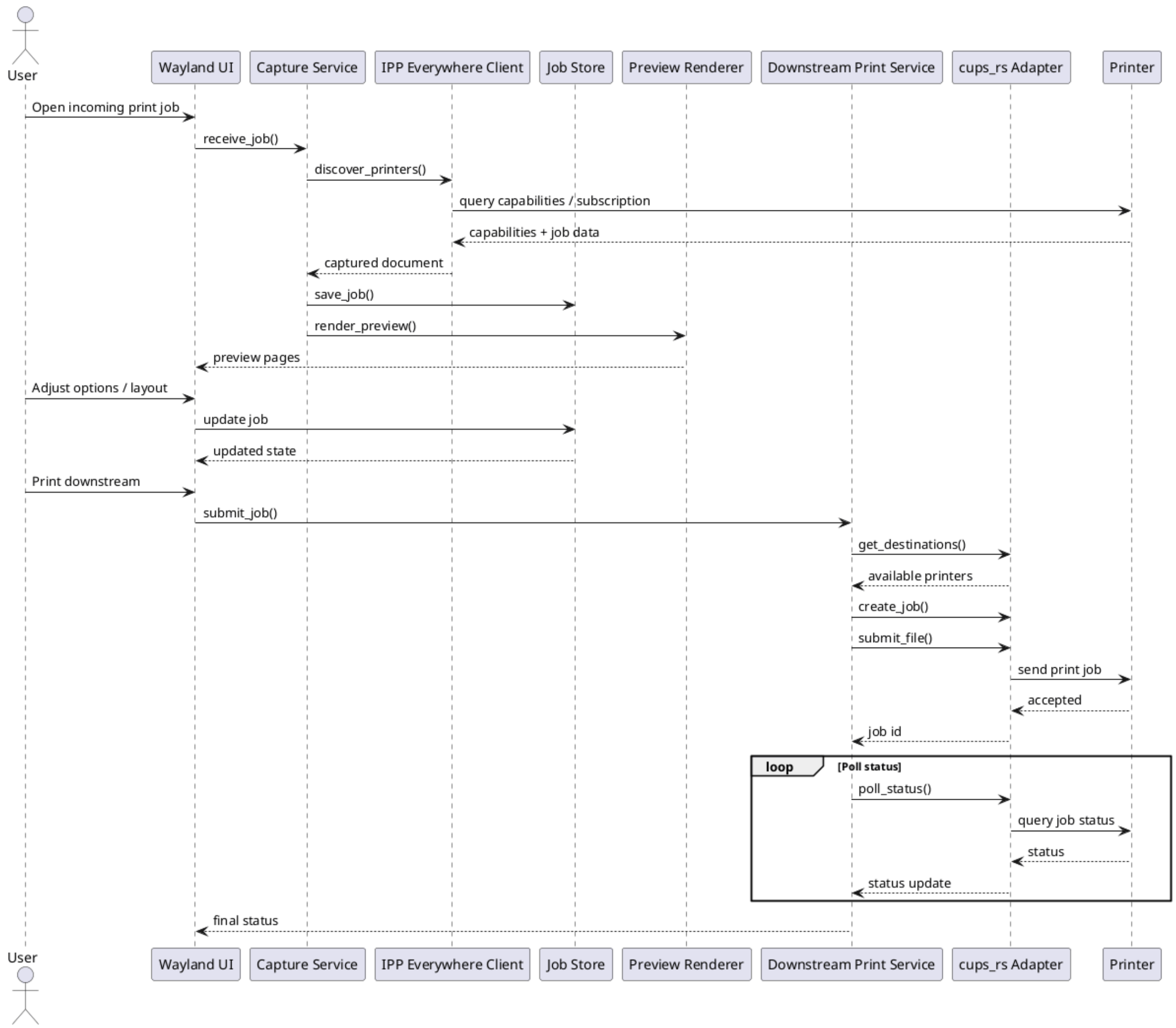

Figure C.3.: *Boomaga-IPP Virtual Printer* System Sequence Diagram.

Appendix C.4. Use Case Diagram

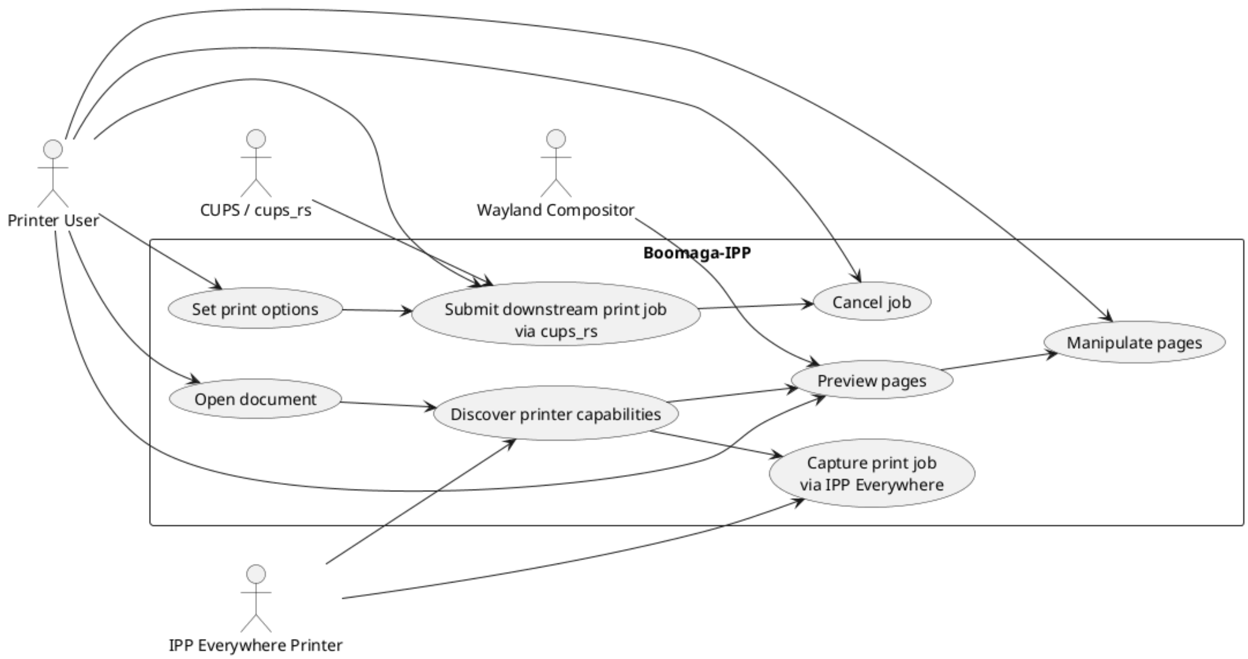

Figure C.4.: *Boomaga-IPP* Use Case.

Appendix D.

To Be Determined (TBD) List

TBD-UIS-001 Additional information to be included on the About *Boomaga-IPP* pop-up menu. See Section 5.5.1, About Boomaga-IPP.